

1 9702/1/M/J/02/Q38

Which set of radioactive emissions corresponds to the descriptions given in the table headings?

	high-speed electrons	high-speed helium nuclei	high-frequency photons
A	α	β	γ
B	α	γ	β
C	β	α	γ
D	β	γ	α

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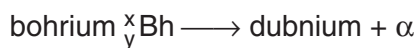
The nucleus of one of the isotopes of nickel is represented by $_{28}\text{Ni}$.

Which line in the table correctly describes a neutral atom of this isotope?

	number of protons	number of neutrons	number of orbital electrons
A	28	32	28
B	28	60	28
C	60	28	28
D	60	32	32

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A nucleus of bohrium ${}^x_y\text{Bh}$ decays to mendelevium ${}^{255}_{101}\text{Md}$ by a sequence of three α -particle emissions.



How many neutrons are there in a nucleus of ${}^X_Y\text{Bh}$?

- A** 267 **C** 160 **B** 261 **D** 154

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The numbers of protons, neutrons and nucleons in three nuclei are shown.

nucleus	number of protons	number of neutrons	number of nucleons
X	15	16	3
Y	15	17	3
Z	16	16	3

Which nuclei are isotopes of the same element?

- A** X and Y **B** X and Z **C** Y and Z **D** none of them

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In an experiment to investigate the nature of the atom, a very thin gold film was bombarded with α -particles.

What pattern of deflection of the α -particles was observed?

- A** A few α -particles were deflected through angles greater than a right angle.
- B** All α -particles were deflected from their original path.
- C** Most α -particles were deflected through angles greater than a right angle.
- D** No α -particle was deflected through an angle greater than a right angle.

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When a nucleus of $^{238}_{92}\text{U}$ absorbs a slow neutron it subsequently emits two β -particles. What is the resulting nucleus?

- A** $^{240}_{93}\text{Np}$ **B** $^{240}_{91}\text{Pa}$ **C** $^{239}_{94}\text{Pu}$ **D** $^{239}_{90}\text{Th}$

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In what way do the atoms of the isotopes $^{12}_6\text{C}$, $^{13}_6\text{C}$ and $^{14}_6\text{C}$ differ?

- A** different charge **C** different numbers of neutrons
- B** different numbers of electrons **D** different numbers of protons

8 9702/01/M/J/03/Q39

Strontium- 90 ($^{90}_{38}\text{Sr}$) is radioactive and emits β -particles.

Which equation could represent this nuclear decay?

- A** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{39}\text{Sr} + ^0_{-1}\beta$ **C** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{37}\text{Rb} + ^0_1\beta$
- B** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{39}\text{Y} + ^0_{-1}\beta$ **D** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{37}\text{Sr} + ^0_1\beta$

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Protons and neutrons are thought to consist of smaller particles called quarks.

The 'up' quark has a charge of $\frac{2}{3}e$: a 'down' quark has a charge of $-\frac{1}{3}e$, where e is the elementary charge ($+1.6 \times 10^{-19}\text{C}$).

How many up quarks and down quarks must a proton contain?

	up quarks	down quarks
A	0	3
B	1	1
C	1	2
D	2	1

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Which are the correct descriptions of a γ -ray and a β -particle?

	γ -ray	β -particle
A	high-speed electron	electromagnetic radiation
B	electromagnetic radiation	Helium-4 nucleus
C	electromagnetic radiation	high-speed electron
D	high-speed electron	Helium-4 nucleus

11 9702/01/O/N/03/Q39

A certain nuclide, Uranium-235, has nucleon number 235, proton number 92 and neutron number 143. Data on four other nuclides are given below.

Which is an isotope of Uranium-235?

	nucleon number	proton number	neutron number
A	235	91	144
B	236	92	144
C	237	94	143
D	238	95	143

12 9702/01/O/N/03/Q40

A nickel nucleus $^{59}_{28}\text{Ni}$ can be transformed by a process termed K-capture. In this process the nucleus absorbs an orbital electron.

If no other process is involved, what is the resulting nucleus?

- A** $^{58}_{28}\text{Ni}$ **B** $^{58}_{27}\text{Co}$ **C** $^{59}_{27}\text{Co}$ **D** $^{59}_{29}\text{Cu}$

13 9702/01/M/J/04/Q38

A nucleus of the nuclide $^{241}_{94}\text{Pu}$ decays by emission of a β -particle followed by the emission of an α -particle. Which of the nuclides shown is formed?

- A** $^{239}_{93}\text{Np}$ **B** $^{239}_{91}\text{Pa}$ **C** $^{237}_{93}\text{Np}$ **D** $^{237}_{92}\text{U}$

14 9702/01/M/J/04/Q40

Isotopes of a given element all have the same

- A** charge/mass ratio. **C** nucleon number.
B neutron number. **D** proton number.

15 9702/01/O/N/04/Q38

The symbol $^{77}_{32}\text{Ge}$ represents a nuclide of germanium that decays to a nuclide of arsenic (As) by emitting a β -particle.

What is the symbol of this arsenic nuclide?

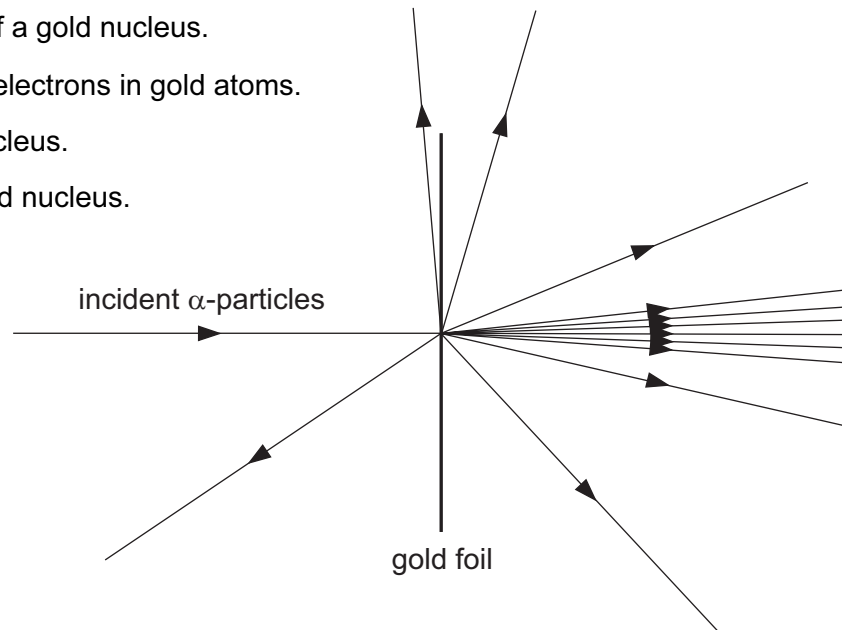
- A** $^{76}_{32}\text{As}$ **B** $^{78}_{32}\text{As}$ **C** $^{78}_{31}\text{As}$ **D** $^{77}_{33}\text{As}$

16 9702/01/M/J/04/Q39*

A thin gold foil is bombarded with α -particles as shown.

The results of this experiment provide information about the

- A binding energy of a gold nucleus.
- B energy levels of electrons in gold atoms.
- C size of a gold nucleus.
- D structure of a gold nucleus.



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The table shows three properties of different types of ionising radiation.

	X	Y	Z
charge	0	$-1e$	$+2e$
mass	0	$\frac{1}{1840}u$	$4u$
speed	c	$\sim 0.9c$	$\sim 0.1c$

What are the radiations X, Y and Z?

	X	Y	Z
A	alpha	beta	X-rays
B	gamma	alpha	beta
C	gamma	beta	alpha
D	X-rays	alpha	beta

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Which conclusion can be drawn from the results of the experiment showing the scattering of α -particles by gold foil?

- A Electrons orbit the atomic nucleus in well-defined paths.
- B Nuclei of different isotopes contain different numbers of neutrons.
- C The atomic nucleus contains protons and neutrons.
- D The nucleus is very small compared with the size of the atom.

19 9702/01/M/J/09/Q28

Which two nuclei contain the same number of neutrons?

- A** $^{12}_6\text{C}$ and $^{14}_6\text{C}$ **C** $^{23}_{11}\text{Na}$ and $^{24}_{12}\text{Mg}$
B $^{16}_7\text{N}$ and $^{15}_8\text{O}$ **D** $^{32}_{14}\text{Si}$ and $^{32}_{15}\text{P}$

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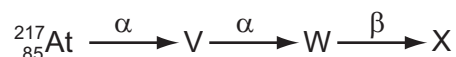
A student conducts an experiment using an α -particle source.

When considering safety precautions, what can be assumed to be the maximum range of α -particles in air?

- A** between 0 and 5 mm **C** between 200 mm and 500 mm
B between 5 mm and 200 mm **D** between 500 mm and 1000 mm

21 9702/01/M/J/05/Q40*

The following represents a sequence of radioactive decays involving two α -particles and one β -particle.



What is the nuclide X?

- A** $^{213}_{85}\text{At}$ **B** $^{215}_{77}\text{Ir}$ **C** $^{209}_{82}\text{Pb}$ **D** $^{217}_{81}\text{Tl}$

22 9702/01/O/N/05/Q38*

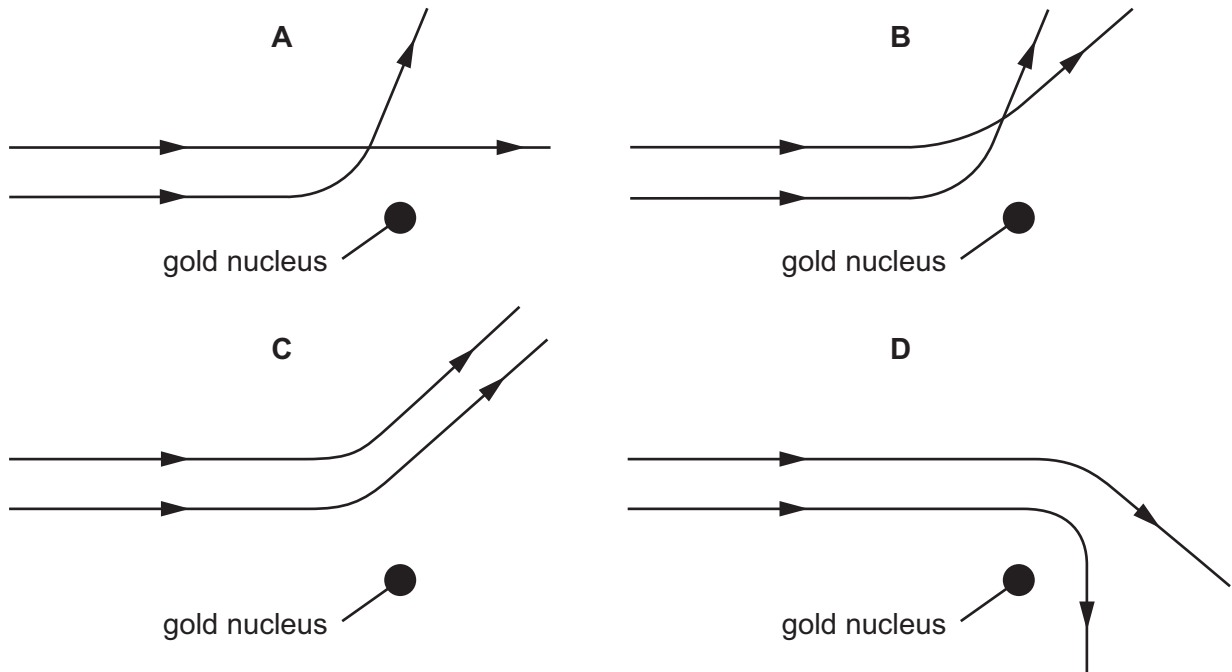
An atomic nucleus emits a β -particle.

What change does this cause to the proton and nucleon numbers of the nucleus?

	proton number	nucleon number
A	−1	+1
B	0	−1
C	+1	−1
D	+1	0

23 9702/01/O/N/05/Q39*

Two α -particles with equal energies are fired towards the nucleus of a gold atom. Which diagram best represents their paths?



24 9702/01/O/N/05/Q40

A nuclear reaction is represented by the equation ${}^{16}_8\text{O} + {}^4_2\text{He} \rightarrow {}^{19}_9\text{F} + \text{X}$. What is particle X?

- A an α -particle B a β -particle C a neutron D a proton

25 9702/01/M/J/06/Q38

What is a correct order of magnitude estimate for the diameter of a typical atomic nucleus?

- A 10^{-14} m B 10^{-18} m C 10^{-22} m D 10^{-26} m

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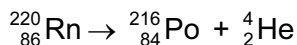
The decay of a nucleus of neptunium is accompanied by the emission of a β -particle and γ -radiation.

What effect (if any) does this decay have on the proton number and the nucleon number of the nucleus?

	proton number	nucleon number
A	increases	decreases
B	decreases	increases
C	unchanged	decreases
D	increases	unchanged

27 9702/01/M/J/06/Q40

Radon-220 is radioactive and decays to Polonium-216 with the emission of an α -particle. The equation for the radioactive decay is shown.



How many neutrons are in the radon and polonium nuclei?

	Rn	Po
A	86	84
B	134	132
C	220	212
D	220	216

28 9702/01/O/N/06/Q38*

Which statement concerning α -particles is correct?

- A** An α -particle has charge $+4e$.
- B** An α -particle is a helium atom.
- C** When α -particles travel through air, they cause ionisation.
- D** When α -particles travel through a sheet of gold foil, they make the gold radioactive.

29 9702/01/O/N/06/Q39

Where are electrons, neutrons and protons found in an atom?

	electrons	neutrons	protons
A	in the nucleus	in the nucleus	orbiting the nucleus
B	in the nucleus	orbiting the nucleus	in the nucleus
C	orbiting the nucleus	in the nucleus	orbiting the nucleus
D	orbiting the nucleus	in the nucleus	in the nucleus

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Radon ${}_{86}^{222}\text{Rn}$ decays by α - and β -emission to bismuth ${}_{83}^{214}\text{Bi}$.

For the decay of each nucleus of radon, how many α - and β -particles are emitted?

	α -particles	β -particles
A	1	1
B	2	1
C	1	2
D	2	2

31 9702/01/M/J/07/Q38

A detector is exposed to a radioactive source. Fluctuations in the count-rate are observed. What do these fluctuations indicate about radioactive decay?

- A** It is random.
- B** It is spontaneous.
- C** It is exponential.
- D** It is non-linear.

32 9702/01/M/J/07/Q39

The symbol ${}^{77}_{32}\text{Ge}$ represents a nucleus of germanium that decays to a nucleus of arsenic by emitting a β -particle.

What is the symbol of this arsenic nucleus?

- A ${}^{76}_{32}\text{As}$ B ${}^{78}_{32}\text{As}$ C ${}^{78}_{31}\text{As}$ D ${}^{77}_{33}\text{As}$

33 9702/01/M/J/07/Q40

Each of the nuclei below is accelerated from rest through the same potential difference.

Which one completes the acceleration with the **lowest** speed?

- A ${}^1_1\text{H}$ B ${}^4_2\text{He}$ C ${}^7_3\text{Li}$ D ${}^9_4\text{Be}$

34 9702/01/O/N/07/Q36

How is it possible to distinguish between the isotopes of uranium?

- A Their nuclei have different charge and different mass, and they emit different particles when they decay.
 B Their nuclei have different charge but the same mass.
 C Their nuclei have the same charge but different mass.
 D Their nuclei have the same charge and mass, but they emit different particles when they decay.

35 9702/01/O/N/07/Q37*

What is **not** conserved in nuclear processes?

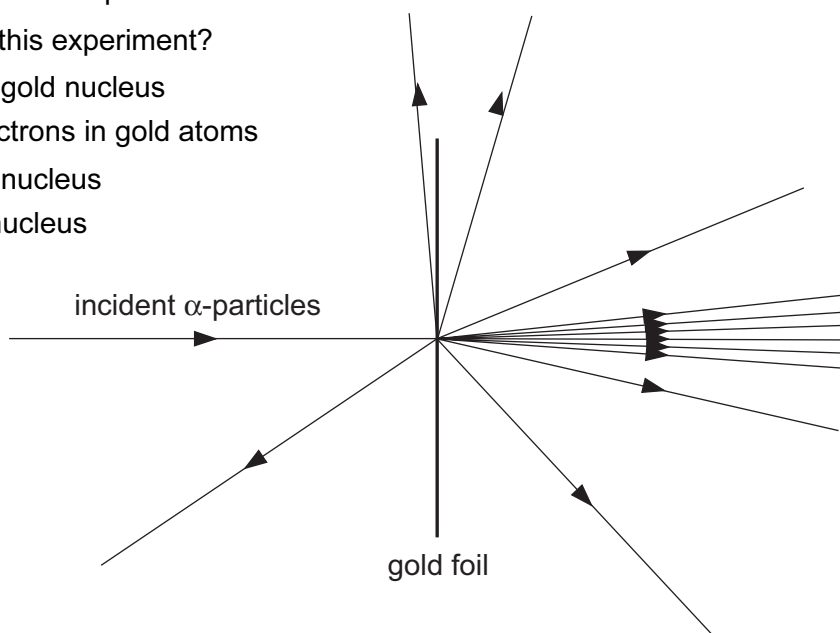
- A energy and mass together C neutron number
 B nucleon number D charge

36 9702/01/O/N/07/Q38

A thin gold foil is bombarded with α -particles as shown.

What can be deduced from this experiment?

- A the binding energy of a gold nucleus
 B the energy levels of electrons in gold atoms
 C the small size of a gold nucleus
 D the structure of a gold nucleus



37 9702/01/O/N/07/Q39

A zirconium nucleus, $^{100}_{40}\text{Zr}$, is a β -emitter. The product nucleus is also a β -emitter.

What is the final resulting nucleus of these two decays?

A $^{100}_{38}\text{Sr}$

B $^{100}_{42}\text{Mo}$

C $^{98}_{40}\text{Zr}$

D $^{102}_{40}\text{Zr}$

38 9702/01/O/N/07/Q40

The following particles are each accelerated from rest through the same potential difference.

Which one completes the acceleration with the **greatest** momentum?

A α -particle

B electron

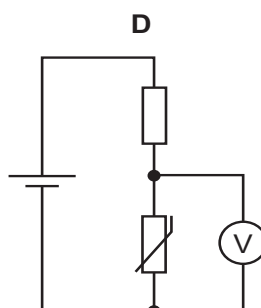
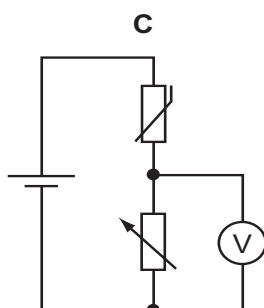
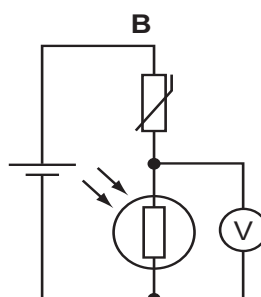
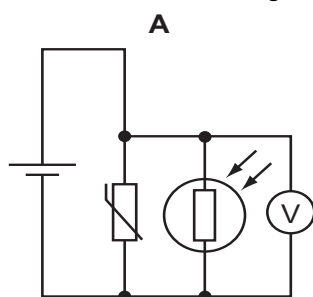
C neutron

D proton

39 9702/01/M/J/08/Q36

A thermistor and another component are connected to a constant voltage supply. A voltmeter is connected across one of the components. The temperature of the thermistor is then reduced but no other changes are made.

In which circuit will the voltmeter reading increase?



40 9702/01/O/N/08/Q38

Which conclusion can be drawn from the results of the experiment showing the scattering of α -particles by gold foil?

A Electrons orbit the atomic nucleus in well-defined paths.

B Nuclei of different isotopes contain different numbers of neutrons.

C The atomic nucleus contains protons and neutrons.

D The nucleus is very small compared with the size of the atom.

41 9702/01/O/N/08/Q39

A nucleus Q has the notation ^y_xQ .

Which of the following is an isotope of Q?

A $^{y-1}_x\text{Q}$

B $^y_{x-1}\text{Q}$

C $^y_{x+1}\text{Q}$

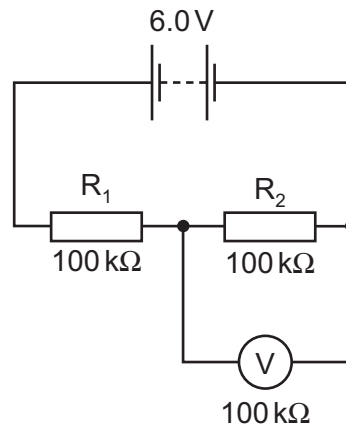
D $^{y-1}_{x+1}\text{Q}$

42 9702/01/M/J/08/Q37

In the circuit shown, the 6.0 V battery has negligible internal resistance. Resistors R_1 and R_2 and the voltmeter have resistance $100\text{ k}\Omega$.

What is the current in the resistor R_2 ?

- A $20\text{ }\mu\text{A}$
- B $30\text{ }\mu\text{A}$
- C $40\text{ }\mu\text{A}$
- D $60\text{ }\mu\text{A}$



43 9702/01/M/J/08/Q39

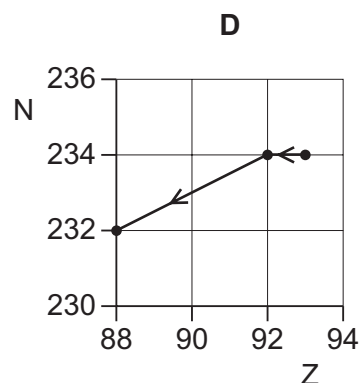
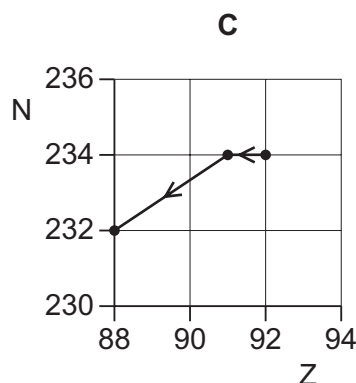
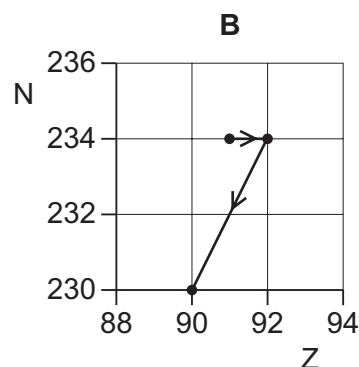
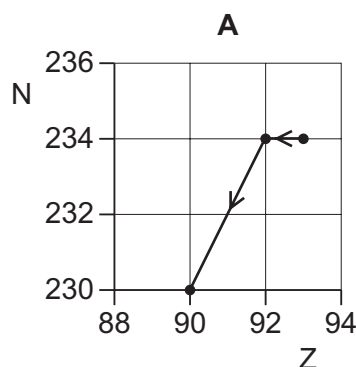
What is the approximate mass of a nucleus of uranium?

- A 10^{-15} kg
- B 10^{-20} kg
- C 10^{-25} kg
- D 10^{-30} kg

44 9702/01/M/J/08/Q40**

A radioactive nucleus is formed by β -decay. This nucleus then decays by α -emission.

Which graph of proton number Z plotted against nucleon number N shows the β -decay followed by the α -emission?



45 9702/01/O/N/08/Q40

A ${}^{238}_{92}\text{U}$ nucleus decays in two stages to a ${}^{234}_{91}\text{Pa}$ nucleus.

What was emitted in these two stages?

- A $\alpha + \beta$
- B $\alpha + \gamma$
- C $\beta + \beta$
- D $\beta + \gamma$

46 9702/01/M/J/09/Q36

How do the nucleon (mass) number and proton (atomic) number of two isotopes of an element compare?

	nucleon number	proton number
A	different	different
B	different	same
C	same	different
D	same	same

47 9702/01/M/J/09/Q37

Nuclear decay is both spontaneous and random.

When the count rate of a radioactive isotope is measured, the readings fluctuate.

Which row describes what the fluctuations demonstrate?

	spontaneous nature	random nature
A	no	no
B	no	yes
C	yes	no
D	yes	yes

48 9702/01/M/J/09/Q38

Which two nuclei contain the same number of neutrons?

- A** $^{12}_6\text{C}$ and $^{14}_6\text{C}$ **C** $^{23}_{11}\text{Na}$ and $^{24}_{12}\text{Mg}$ **B** $^{16}_7\text{N}$ and $^{15}_8\text{O}$ **D** $^{32}_{14}\text{Si}$ and $^{32}_{15}\text{P}$

49 9702/01/M/J/09/Q39

The calcium nuclide $^{42}_{20}\text{Ca}$ is formed by beta decay.

What are the nucleon (mass) number and proton (atomic) number of the unstable nuclide that underwent beta decay to form the calcium nuclide?

	nucleon number	proton number
A	41	19
B	41	21
C	42	19
D	42	21

50 9702/01/M/J/09/Q40

When boron-11 ($^{11}_5\text{B}$) is bombarded with α -particles, a new nucleus is formed and a neutron is released.

Which nuclear equation could represent this reaction?

- A** $^{11}_5\text{B} + ^1_2\text{He} \rightarrow ^{11}_6\text{C} + ^1_0\text{n}$ **C** $^{11}_5\text{B} + ^4_2\text{He} \rightarrow ^{14}_6\text{C} + ^1_1\text{n}$
B $^{11}_5\text{B} + ^2_2\text{He} \rightarrow ^{12}_7\text{N} + ^1_0\text{n}$ **D** $^{11}_5\text{B} + ^4_2\text{He} \rightarrow ^{14}_7\text{N} + ^1_0\text{n}$

51 9702/12/O/N/09/Q36

The gold nucleus $^{185}_{79}\text{Au}$ undergoes alpha decay.

What are the nucleon (mass) number and proton (atomic) number of the nucleus formed by this decay?

	nucleon number	proton number
A	183	79
B	183	77
C	181	77
D	181	75

52 9702/12/O/N/09/Q37

The nuclei of the isotopes of an element all contain the same number of a certain particle.

What is this particle?

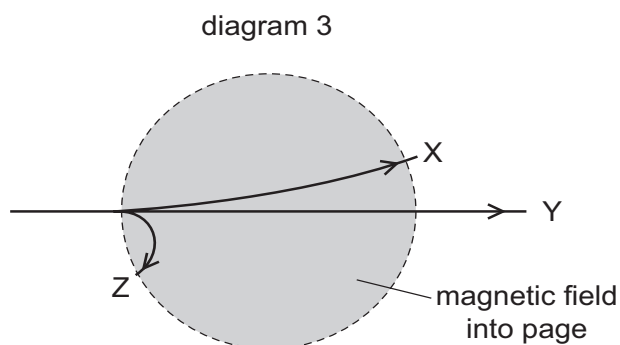
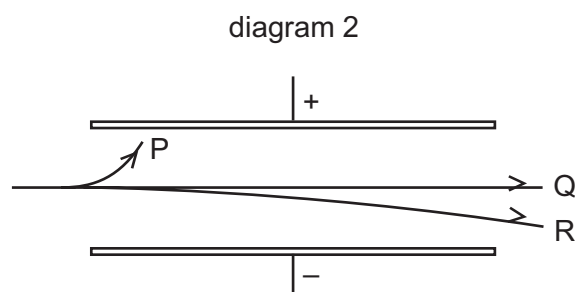
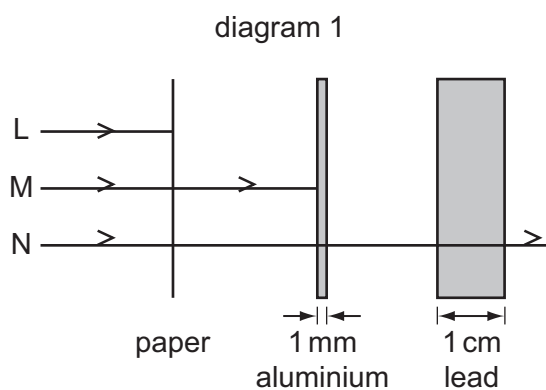
- A** electron **B** neutron **C** nucleon **D** proton

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Alpha, beta and gamma radiations

- 1 are absorbed to different extents in solids,
- 2 behave differently in an electric field,
- 3 behave differently in a magnetic field.

The diagrams illustrate these behaviours.



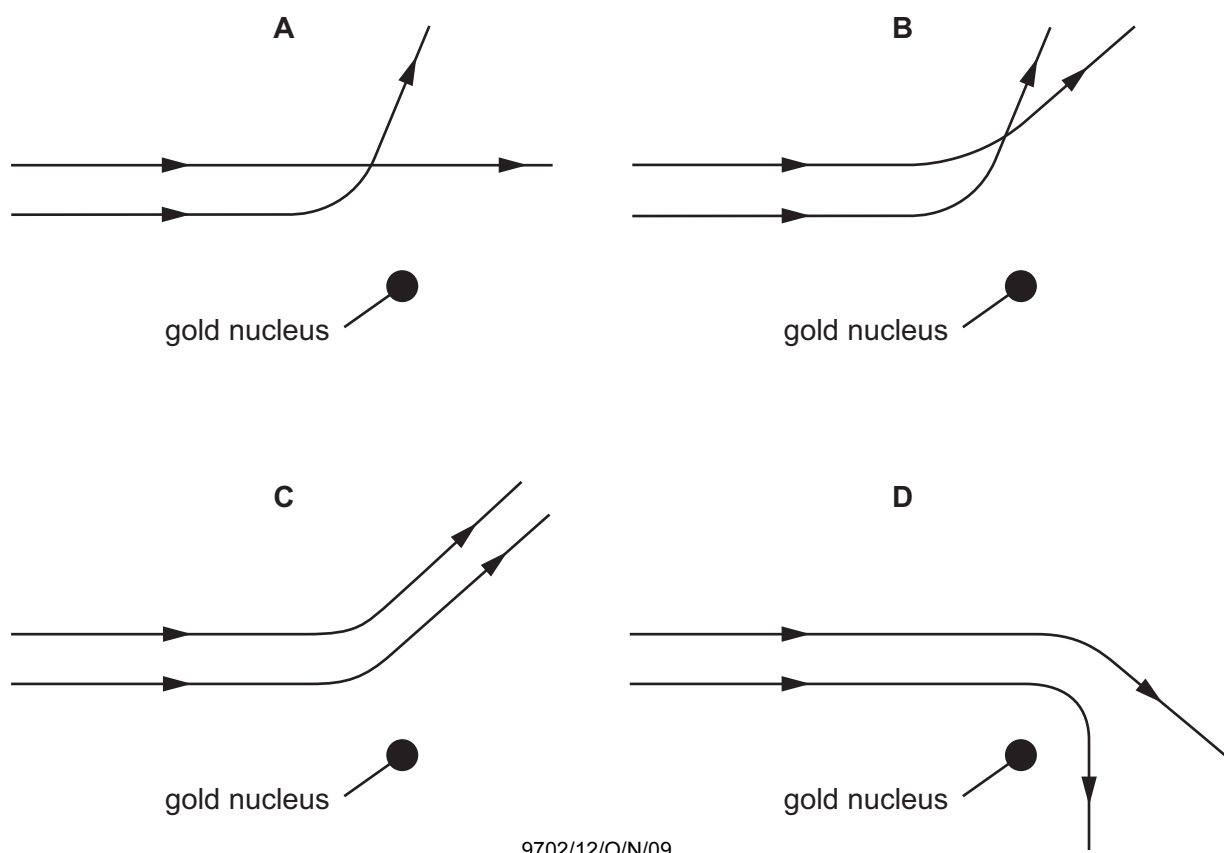
Which three labels on these diagrams refer to the **same** kind of radiation?

- A** L, P, X **B** L, P, Z **C** M, P, Z **D** N, Q, X

54 9702/01/O/N/05/Q39

Two α -particles with equal energies are fired towards the nucleus of a gold atom.

Which diagram best represents their paths?



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The grid shows a number of nuclides arranged according to the number of protons and the number of neutrons in each.

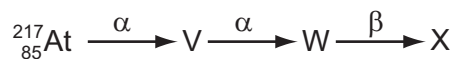
A nucleus of the nuclide ${}^8_3\text{Li}$ decays by emitting a β -particle.

What is the resulting nuclide?

number of protons	4					A	B	
	3				${}^6_3\text{Li}$	${}^7_3\text{Li}$	${}^8_3\text{Li}$	
	2		${}^3_2\text{He}$	${}^4_2\text{He}$			C	D
	1	${}^1_1\text{H}$	${}^2_1\text{H}$					
		0	1	2	3	4	5	6
		number of neutrons						

56 9702/12/M/J/10/Q38

The following represents a sequence of radioactive decays involving two α -particles and one β -particle.



What is the nuclide X?

- A ${}_{85}^{213}\text{At}$ B ${}_{77}^{215}\text{Ir}$ C ${}_{82}^{209}\text{Pb}$ D ${}_{81}^{217}\text{Tl}$

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What are the correct descriptions of a γ -ray and a β -particle?

	γ -ray	β -particle
A	high-speed electron	electromagnetic radiation
B	electromagnetic radiation	helium-4 nucleus
C	electromagnetic radiation	high-speed electron
D	high-speed electron	helium-4 nucleus

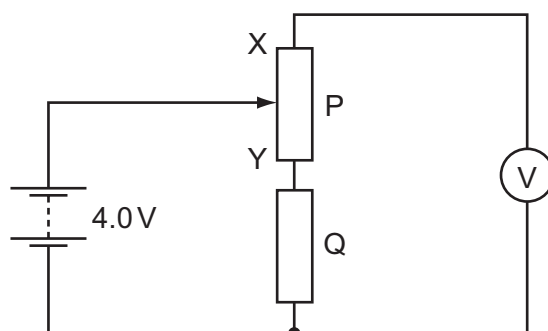
58 9702/12/M/J/10/Q40

What is **not** conserved in nuclear processes?

- A charge C the total number of neutrons
B momentum D the total number of nucleons

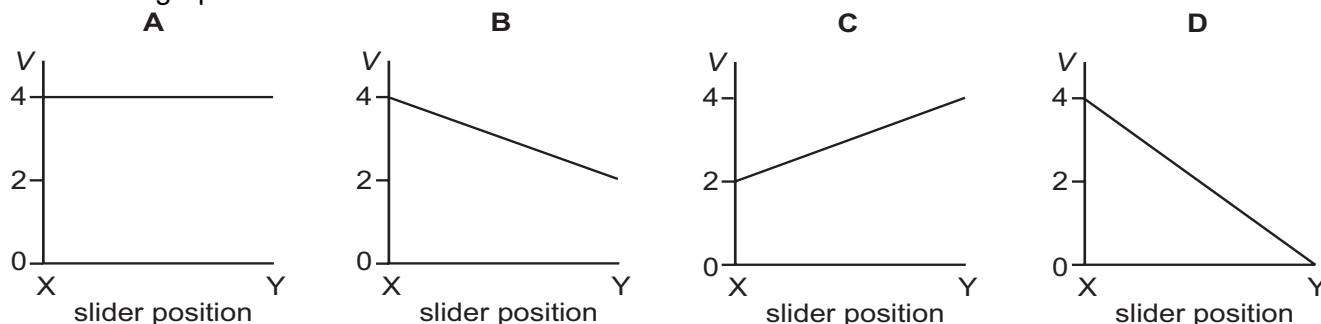
59 9702/11/O/N/10/Q37

In the circuit below, P is a potentiometer of total resistance $10\ \Omega$ and Q is a fixed resistor of resistance $10\ \Omega$. The battery has an e.m.f. of $4.0\ \text{V}$ and negligible internal resistance. The voltmeter has a very high resistance.



The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading V is plotted against slider position.

Which graph would be obtained?



60 9702/11/O/N/10/Q38

Uranium-235 may be represented by the symbol $^{235}_{92}\text{U}$.

Which row shows the numbers of nucleons, protons and neutrons in a $^{235}_{92}\text{U}$ nucleus?

	nucleons	protons	neutrons
A	92	235	143
B	143	92	235
C	235	92	143
D	235	143	92

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When a magnesium nucleus $^{25}_{12}\text{Mg}$ is hit by a gamma ray, a sodium nucleus $^{24}_{11}\text{Na}$ is formed and another particle is emitted.

What are the nucleon number (mass number) and proton number (atomic number) of the other particle produced in this nuclear reaction?

	nucleon number	proton number
A	0	-1
B	0	1
C	1	-1
D	1	1

62 9702/11/O/N/10/Q40

Which nuclear equation shows the beta decay of a nucleus of argon (Ar) into potassium (K)?



63 9702/12/O/N/10/Q37

Three resistors, with resistances R_1 , R_2 and R_3 , are connected in series and are found to have a combined resistance of $500\ \Omega$. When connected in parallel, the combined resistance is found to be $50\ \Omega$.

Which values will correspond to these results?

	R_1/Ω	R_2/Ω	R_3/Ω
A	160	160	80
B	200	200	100
C	225	225	50
D	230	230	40

64 9702/12/O/N/10/Q38

In the Rutherford scattering experiment, α -particles were fired at a thin gold foil. A small proportion of the α -particles were deflected through large angles.

Which statement gives the correct conclusion that could be drawn directly from these results?

- A** The atom is made up of electrons, protons and neutrons.
- B** The nucleus is at the centre of the atom.
- C** The nucleus is made up of protons and neutrons.
- D** The atom contains a very small, charged nucleus.

65 9702/12/O/N/10/Q39

Which statement about the nuclei of the atoms of an element is correct?

- A** Every nucleus of an element contains an equal number of neutrons and protons.
- B** Every nucleus of an element contains the same number of neutrons as all others of that element, but the number of protons may differ.
- C** Every nucleus of an element contains the same number of protons as all others of that element, but the number of neutrons may differ.
- D** The number of protons in a nucleus differs from isotope to isotope of an element, as do the number of neutrons.

66 9702/12/O/N/10/Q40

A counter recording radioactive decays from a radioactive source gives the following counts in equal intervals of time.

time / min	counts
0–10	424
10–20	395
20–30	413
30–40	363
40–50	366
50–60	294
60–70	301
70–80	253
80–90	212

What can be deduced from these readings?

- A** that radioactivity is random and that the half-life is 90 minutes
- B** that radioactivity is random and that the half-life is uncertain
- C** that radioactivity is spontaneous and that the half-life is 90 minutes
- D** that radioactivity is spontaneous and that the half-life is uncertain

67 9702/11/M/J/11/Q38

The first artificial radioactive substance was made by bombarding aluminium, $^{27}_{13}\text{Al}$, with α -particles. This produced an unstable isotope of phosphorus, $^{30}_{15}\text{P}$.

What was the by-product of this reaction?

- A** an α -particle **B** a β -particle **C** a γ -ray **D** a neutron

68 9702/11/M/J/11/Q39

Uranium-238, $^{238}_{92}\text{U}$, decays by α -emission into a daughter product which in turn decays by β -emission into a grand-daughter product.

What is the grand-daughter product?

- A** $^{234}_{90}\text{Th}$ **B** $^{234}_{91}\text{Pa}$ **C** $^{234}_{92}\text{U}$ **D** $^{230}_{90}\text{Th}$

69 9702/11/M/J/11/Q40

Which statement about nuclei is correct?

- A** Different isotopic nuclei have different proton numbers.
B For some nuclei, the nucleon number can be less than the proton number.
C In some nuclear processes, mass-energy is not conserved.
D Nucleon numbers of nuclei are unchanged by the emission of β -particles.

70 9702/12/M/J/11/Q39

The uranium nucleus $^{238}_{92}\text{U}$ undergoes α -decay, producing nucleus X.

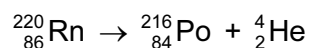
Nucleus X undergoes β -decay, producing nucleus Y.

For nucleus Y, what are the values of the proton number and nucleon number?

	proton number	nucleon number
A	89	234
B	89	236
C	91	234
D	91	236

71 9702/12/M/J/11/Q40

Radon-220 is radioactive and decays to polonium-216 with the emission of an α -particle. The equation for the radioactive decay is shown.



How many neutrons are in the radon and polonium nuclei?

	Rn	Po
A	86	84
B	134	132
C	220	212
D	220	216

72 9702/11/O/N/11/Q40

An atomic nucleus emits a β -particle.

What change does this cause to the proton number and the nucleon number of the nucleus?

	proton number	nucleon number
A	-1	+1
B	0	-1
C	+1	-1
D	+1	0

73 9702/12/O/N/11/Q39

Which statement concerning α -particles is correct?

- A** An α -particle has charge $+4e$.
- B** An α -particle is a helium atom.
- C** When α -particles travel through air, they cause ionisation.
- D** When α -particles travel through a sheet of gold foil, they make the gold radioactive.

74 9702/12/O/N/11/Q40

A nucleus of the nuclide ${}_{94}^{241}\text{Pu}$ decays by emission of a β -particle followed by the emission of an α -particle.

Which nucleus is formed?

- A** ${}_{93}^{239}\text{Np}$ **B** ${}_{91}^{239}\text{Pa}$ **C** ${}_{93}^{237}\text{Np}$ **D** ${}_{92}^{237}\text{U}$

75 9702/12/M/J/12/Q39

Nuclear decay is both spontaneous and random in nature.

Which row gives the correct experimental evidence for these properties?

	spontaneous nature of decay	random nature of decay
A	the decay rate is not affected by pressure	the decay rate is not affected by temperature
B	the decay rate is not affected by pressure	the rate at which radiation is received at a counter fluctuates
C	the decay rate is not affected by temperature	the decay rate is not affected by pressure
D	the rate at which radiation is received at a counter fluctuates	the decay rate is not affected by pressure

76 9702/12/M/J/12/Q40

Radon $^{222}_{86}\text{Rn}$ is the start of a decay chain that forms bismuth $^{214}_{83}\text{Bi}$ by alpha and beta emission.

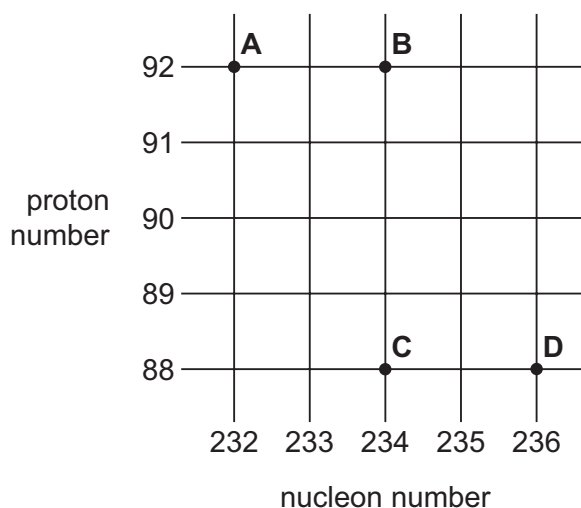
For the decay of each nucleus of radon, how many α -particles and β -particles are emitted?

	α -particles	β -particles
A	1	1
B	2	1
C	1	2
D	2	2

77 9702/13/M/J/12/Q39

Thorium-234 ($^{234}_{90}\text{Th}$) decays by β -emission into a daughter product which in turn decays by further β -emission into a granddaughter product.

Which letter in the diagram represents the granddaughter product?



78 9702/13/M/J/12/Q40

An experiment in which α -particles were deflected by a gold foil produced new insights into the structure of the atom.

Which conclusion can be drawn from the results of the experiment?

- A** Atomic nuclei occupy a very small fraction of the volume of an atom.
- B** Electrons orbit the atomic nucleus.
- C** Some atoms of the same element contain different numbers of neutrons.
- D** The atomic nucleus contains protons and neutrons.

79 9702/12/O/N/12/Q39

A material contains a radioactive isotope that disintegrates solely by the emission of α -particles at a rate of 100 s^{-1} .

Which statement about this material is correct?

- A** The number of atoms in the material diminishes at a rate of 100 s^{-1} .
- B** The number of neutrons in the material diminishes at a rate of 100 s^{-1} .
- C** The number of nucleons in the material diminishes at a rate of 400 s^{-1} .
- D** The number of protons in the material diminishes at a rate of 100 s^{-1} .

80 9702/12/O/N/12/Q40

A different nucleus can be formed by bombarding a stable nucleus with an energetic α -particle.

$^{23}_{11}\text{Na}$ is bombarded with an energetic α -particle.

What could be the products of this nuclear reaction?

- A** $^{25}_{10}\text{Ne}$ + neutron **B** $^{25}_{11}\text{Na}$ + proton **C** $^{26}_{12}\text{Mg}$ + β **D** $^{27}_{13}\text{Al}$ + γ

81 9702/13/O/N/12/Q38

A nuclear isotope emits radiation which is detected by a Geiger-Müller tube held at a distance of about 10 cm from the radioactive source. The radiation is stopped completely by a 2 mm thick sheet of lead.

What can be deduced from this information about the emission from the isotope?

- A** It could be alpha and beta radiation, but not gamma radiation.
- B** It could be alpha and gamma radiation, but not beta radiation.
- C** It could be beta and gamma radiation, but not alpha radiation.
- D** It could be alpha, beta and gamma radiation.

82 9702/13/O/N/12/Q39

What remains constant during β -emission from a number of identical nuclei in a substance?

- A** energy of the β -particles **C** nucleon number of the nuclei
- B** neutron number of the nuclei **D** proton number of the nuclei

83 9702/11/M/J/13/Q38

A nickel nucleus $^{59}_{28}\text{Ni}$ can be transformed by a process termed K-capture. In this process the nucleus absorbs an orbital electron.

If no other process is involved, what is the resulting nucleus?

- A $^{58}_{28}\text{Ni}$ B $^{58}_{27}\text{Co}$ C $^{59}_{27}\text{Co}$ D $^{59}_{29}\text{Cu}$

84 9702/11/M/J/13/Q39

It was once thought that the mass of an atom is spread uniformly through the volume of the atom. When α -particles are directed at a piece of gold foil, the results led scientists to believe instead that nearly all the mass of the gold atom is concentrated at a point inside the atom.

Which effect is possible **only** if nearly all the mass of the gold atom is concentrated at a point?

- A a few α -particles bounce back
B most α -particles are only slightly deflected
C some α -particles pass through without any deflection
D some α -particles are absorbed

85 9702/13/M/J/13/Q38

Scientists investigating the count rate from a radioactive source observed that the count rate fluctuates.

What do these fluctuations imply about the nature of radioactive decay?

- A It involves atomic nuclei. B It is predictable. C It is random. D It is spontaneous.

86 9702/11/M/J/13/Q40

Which pair of nuclei are isotopes of one another?

	nucleon number	number of neutrons
A	186	112
	180	118
B	186	112
	182	108
C	184	110
	187	110
D	186	110
	186	112

87 9702/12/M/J/13/Q39

What is the approximate mass of an alpha particle?

- A 10^{-28} kg B 10^{-26} kg C 10^{-24} kg D 10^{-22} kg

88 9702/12/M/J/13/Q40

An actinium nucleus has a nucleon number of 227 and a proton number of 89. It decays to form a radium nucleus, emitting a beta particle and an alpha particle in the process.

What are the nucleon number and the proton number of this radium nucleus?

	nucleon number	proton number
A	223	87
B	223	88
C	224	87
D	225	86

89 9702/13/M/J/13/Q39

The decay of a nucleus of neptunium is accompanied by the emission of a β -particle and γ -radiation.

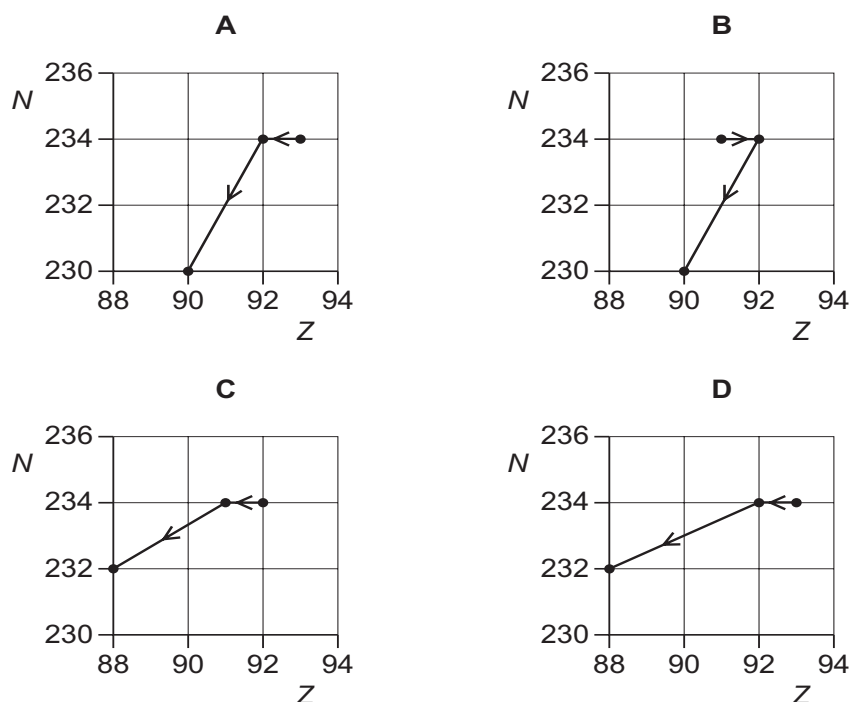
What effect (if any) does this decay have on the proton number and on the nucleon number of the nucleus?

	proton number	nucleon number
A	increases	decreases
B	decreases	increases
C	unchanged	decreases
D	increases	unchanged

90 9702/13/M/J/13/Q40

A radioactive nucleus is formed by β -decay. This nucleus then decays by α -emission.

Which graph of nucleon number N plotted against proton number Z shows the β -decay followed by the α -emission?



91 9702/12/O/N/13/Q39

When α -particles are fired at a thin metal foil, most of the particles pass straight through but a few are deflected by a large angle.

Which change would increase the **proportion** of α -particles deflected by a large angle?

- A using α -particles with greater kinetic energy
- B using a foil made of a metal with fewer protons in its nuclei
- C using a double thickness foil
- D using an alpha source with a higher activity

92 9702/12/O/N/13/Q40

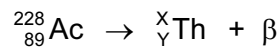
Plutonium-239 ($^{239}_{94}\text{Pu}$) decays by emitting α -radiation.

Which nuclide is formed from one of these decay reactions? (The product nuclides are represented by X.)

- A $^{235}_{92}\text{X}$ B $^{237}_{92}\text{X}$ C $^{239}_{93}\text{X}$ D $^{239}_{95}\text{X}$

93 9702/13/O/N/13/Q39

A nucleus of the nuclide $^{228}_{89}\text{Ac}$ decays by emitting a beta particle. The nuclear equation below represents this decay.



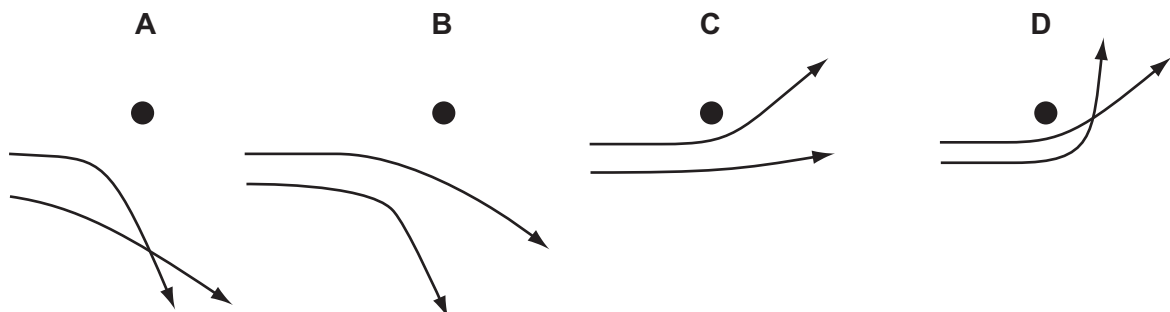
Which pair of values of X and Y is correct?

	X	Y
A	224	87
B	224	89
C	228	88
D	228	90

94 9702/13/O/N/13/Q40

Two α -particles with equal energies are deflected by a large nucleus.

Which diagram best represents their paths?



95 9702/11/M/J/14/Q37

Alpha, beta and gamma radiations have various depths of penetration in matter and different charges.

Which row best summarises the penetration and charge of each radiation?

	alpha	beta	gamma
A	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium no charge	not fully absorbed by several cm of lead no charge
B	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium positive charge	not fully absorbed by several cm of lead no charge
C	absorbed by a sheet of card positive charge	absorbed by several mm of aluminium negative charge	not fully absorbed by several cm of lead no charge
D	absorbed by several mm of aluminium positive charge	not fully absorbed by several cm of lead negative charge	absorbed by a sheet of card no charge

96 9702/11/M/J/14/Q38

In 2002, two-proton radioactive decay of an isotope of iron, ${}^{45}_{26}\text{Fe}$, was observed.

What could be the resulting product?

- A** ${}^{43}_{26}\text{Fe}$ **B** ${}^{43}_{24}\text{Cr}$ **C** ${}^{45}_{24}\text{Cr}$ **D** ${}^{47}_{28}\text{Ni}$

97 9702/11/M/J/14/Q39

U^{++} is a doubly-ionised uranium atom. The uranium atom has a nucleon number of 235 and a proton number of 92.

In a simple model of the atom, how many particles are in this ionised atom?

- A** 235 **B** 325 **C** 327 **D** 329

98 9702/11/M/J/14/Q40

The grid shows a number of nuclides arranged according to the number of protons and the number of neutrons in each.

A nucleus of the nuclide ${}^8_3\text{Li}$ decays by emitting a β -particle.

What is the resulting nuclide?

					A	B	
4							
3				${}^6_3\text{Li}$	${}^7_3\text{Li}$	${}^8_3\text{Li}$	
2		${}^3_2\text{He}$	${}^4_2\text{He}$			C	D
1	${}^1_1\text{H}$	${}^2_1\text{H}$					
	0	1	2	3	4	5	6
	number of neutrons						

99 9702/12/M/J/14/Q38

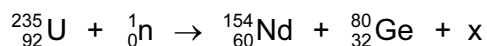
A nucleus X decays into a nucleus Y by emitting an alpha particle followed by two beta particles.

Which statement about this nuclear decay is correct?

- A Beta particle decay occurs when a proton changes into a neutron.
- B Nucleus Y has the same nucleon number as nucleus X.
- C Nucleus Y is an isotope of nucleus X.
- D The total mass of the products is equal to the mass of the initial nucleus X.

100 9702/12/M/J/14/Q39

A slow-moving neutron collides with a nucleus of uranium-235. This results in a nuclear reaction that is represented by the following nuclear equation



where x represents one or more particles.

What does x represent?

- A one neutron
- B two electrons
- C two neutrons
- D two protons

101 9702/12/M/J/14/Q40

The first artificial radioactive substance was made by bombarding aluminium, ${}^{27}_{13}\text{Al}$, with α -particles. This produced an unstable isotope of phosphorus, ${}^{30}_{15}\text{P}$.

What was the by-product of this reaction?

- A an α -particle
- B a β -particle
- C a neutron
- D a proton

102 9702/13/M/J/14/Q39

An isotope of thorium has a nucleon number of 232 and a proton number of 90. It decays to form another isotope of thorium with a nucleon number of 228.

How many alpha particles and beta particles are emitted by a nucleus of thorium during this decay?

	alpha particles	beta particles
A	0	4
B	1	0
C	1	2
D	2	1

103 9702/13/M/J/14/Q40

Four nuclei are represented below.



Which statement about these nuclei is correct?

- A** An uncharged atom of element Q has 24 orbital electrons.
- B** Nucleus M could transform into Q by emitting a beta particle.
- C** Nuclei G and M are isotopes of the same element.
- D** When E absorbs a neutron and then emits an alpha particle, nucleus E transforms into M.

104 9702/11/O/N/14/Q35

Which statement about α -particles is correct?

- A** α -particles emitted from a single radioactive isotope have a continuous distribution of energies.
- B** α -particles have less ionising power than β -particles.
- C** The charge of an α -particle is $+1.60 \times 10^{-19} \text{ C}$.
- D** The speeds of α -particles can be as high as $1.5 \times 10^7 \text{ m s}^{-1}$.

105 9702/11/O/N/14/Q39

The isotope ${}_{86}^{222}\text{Rn}$ decays in a sequence of emissions to form the isotope ${}_{82}^{206}\text{Pb}$. At each stage of the decay sequence, it emits either an α -particle or a β -particle.

What is the number of stages in the decay sequence?

- A** 4 **B** 8 **C** 16 **D** 20

106 9702/12/O/N/14/Q40

What is the approximate mass of a nucleus of uranium?

- A** 10^{-15} kg **B** 10^{-20} kg **C** 10^{-25} kg **D** 10^{-30} kg

107 9702/13/O/N/14/Q38

The nucleus of a radioactive isotope of an element emits an alpha particle. The daughter nucleus then emits a beta particle and then the daughter nucleus of that reaction emits another beta particle.

Which statement describes the final nuclide that is formed?

- A** It is a different isotope of the original element.
B It is a nuclide of a different element of higher proton number.
C It is a nuclide of the same element but with different proton number.
D It is identical to the original nuclide.

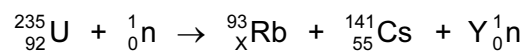
108 9702/13/O/N/14/Q39

A nuclear reaction is shown. ${}^{238}_{92}\text{U} + {}^4_2\text{He} \rightarrow {}^{241}_{94}\text{Pu} + \text{X}$ What is product X?

- A** an alpha particle **B** an electron **C** a neutron **D** a proton

109 9702/13/M/J/15/Q40

The nuclear equation for a fission reaction is shown below.



What are the values of X and Y?

	X	Y
A	37	0
B	37	1
C	37	2
D	38	2

110 9702/11/M/J/15/Q39

When α -particles are directed at gold leaf

- 1 almost all α -particles pass through without deflection,
- 2 a few α -particles are deviated through large angles.

What are the reasons for these effects?

	1	2
A	most α -particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α -particles bounce back from the gold surface
B	most α -particles miss all gold atoms	a few α -particles bounce off gold atoms
C	the gold nucleus is very small so most α -particles miss all nuclei	occasionally the path of an α -particle is close to a nucleus
D	the positive charge in an atom is not concentrated enough to deflect an α -particle	occasionally an α -particle experiences many small deflections in the same direction

111 9702/11/M/J/15/Q40

The nuclide $^{222}_{86}\text{Rn}$ decays in a sequence of stages to form the nuclide $^{206}_{82}\text{Pb}$.

Four of the nuclides formed in the sequence are α -particle emitters. The others are β -particle emitters.

How many nuclides formed in the decay sequence are β -particle emitters?

- A** 2 **B** 4 **C** 8 **D** 12

112 9702/12/M/J/15/Q38

An experiment in which α -particles were deflected by a gold foil produced new insights into the structure of the atom.

Which conclusion can be drawn from the results of the experiment?

- A** Atomic nuclei occupy a very small fraction of the volume of an atom.
B Electrons orbit the atomic nucleus.
C Some atoms of the same element contain different numbers of neutrons.
D The atomic nucleus contains protons and neutrons.

113 9702/12/M/J/15/Q39

A radioactive substance contains a number of identical nuclei that emit β -particles.

Which property of these nuclei remains unaltered by the emission?

- A** charge **B** neutron number **C** nucleon number **D** proton number

114 9702/12/M/J/15/Q40

A uranium-238 nucleus, ${}^{238}_{92}\text{U}$, undergoes nuclear decays to form uranium-234, ${}^{234}_{92}\text{U}$.

Which series of decays could give this result?

- A** emission of four β -particles **C** emission of one α -particle and two β -particles
B emission of four γ -rays **D** emission of two α -particles and eight β -particles

115 9702/13/M/J/15/Q38

Nucleus P decays in two stages to produce nucleus Q.

Which decay sequence will result in the highest number of neutrons in nucleus Q?

- A** an α -particle followed by a β -particle **C** a β -particle followed by another β -particle
B an α -particle followed by a γ -ray **D** a β -particle followed by a γ -ray

116 9702/13/M/J/15/Q39

A neutral atom has a nucleus given by the symbol ${}^{133}_{55}\text{Cs}$.

How many protons, neutrons and electrons are in this atom?

	protons	neutrons	electrons
A	55	78	55
B	55	133	55
C	78	55	78
D	133	55	133

117 9702/12/M/J/17/Q39

A certain nuclide, uranium-235, has nucleon number 235, proton number 92 and neutron number 143. Data on four other nuclides are given below.

Which nuclide is an isotope of uranium-235?

	nucleon number	proton number	neutron number
A	235	91	144
B	236	92	144
C	237	94	143
D	238	95	143

118 9702/12/M/J/17/Q40

During β^- decay, which change takes place to the quark composition of the nucleus that emits the β^- particle, and which other particle is emitted?

	quark change	other particle emitted
A	down to up	antineutrino
B	down to up	neutrino
C	up to down	antineutrino
D	up to down	neutrino

119 9702/11/M/J/17/Q38

The table lists the nucleon number and the proton number of various nuclei. The nuclei are represented by the letters L to T.

nucleus	nucleon number	proton number
L	227	89
M	226	89
N	225	89
O	227	90
P	226	90
Q	225	90
R	227	91
S	226	91
T	225	91

Which row in the following table correctly shows three nuclei of the same element, and three nuclei that have the same number of neutrons?

	same element	same number of neutrons
A	L M N	R P N
B	M P S	R S T
C	O P Q	M P S
D	R P N	O P Q

120 9702/12/M/J/17/Q38

In a television programme to illustrate scientific models, a presenter fires a gun many times at a bale of hay. Two small cannon balls are embedded within the hay some distance apart from each other.

The hay bale measures approximately $2\text{ m} \times 2\text{ m} \times 2\text{ m}$ and the cannon balls are made of iron, approximately spherical, and about 5 cm in diameter.

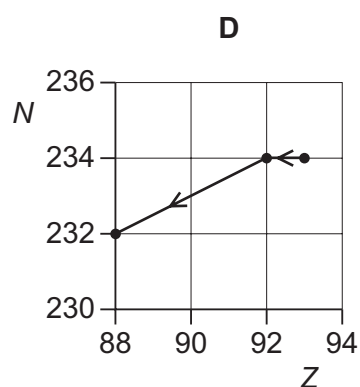
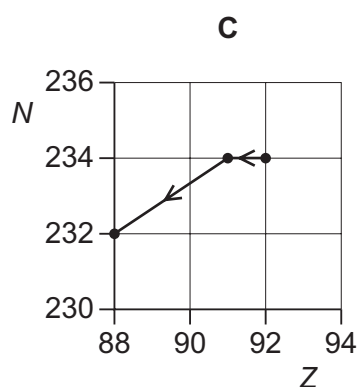
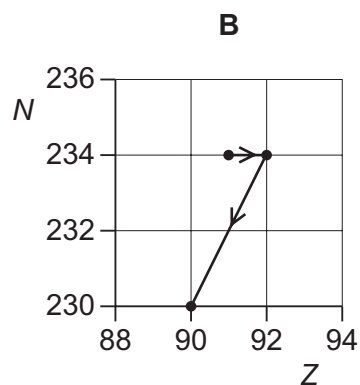
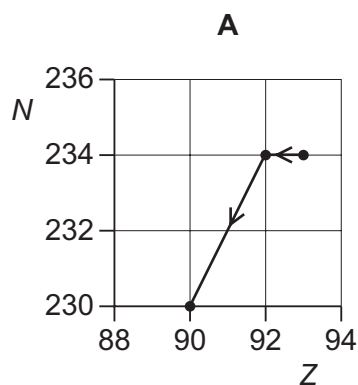
What might the presenter be illustrating?

- A** α -particle scattering
- B** β^- decay
- C** conservation of momentum
- D** double-slit interference

121 9702/01/M/J/08/Q40

A radioactive nucleus is formed by β^- decay. This nucleus then decays by α -emission.

Which graph of nucleon number N plotted against proton number Z shows the β^- decay followed by the α -emission?



122 9702/11/M/J/17/Q40

What are the structures of the proton and of the neutron in terms of quarks?

	proton		neutron	
	up quark	down quark	up quark	down quark
A	1	1	2	2
B	1	2	2	1
C	2	1	1	2
D	2	2	1	1

123 9702/11/O/N/17/Q39

An astatine nucleus has a nucleon number of 218 and a proton number of 85. It decays to form a polonium nucleus, emitting a β^- particle and an α -particle in the process.

What are the nucleon number and the proton number of this polonium nucleus?

	nucleon number	proton number
A	214	83
B	214	84
C	215	83
D	216	82

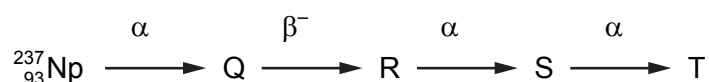
124 9702/11/O/N/17/Q40

What is the quark composition of a hydrogen-3 nucleus, ${}^3_1\text{H}$?

	number of quarks	
	up	down
A	4	5
B	5	4
C	5	7
D	7	5

125 9702/12/O/N/17/Q39

The diagram shows a sequence of radioactive decays involving three α -particles and a β^- particle.



What is nuclide T?

- A** ${}^{225}_{88}\text{Ra}$ **B** ${}^{231}_{88}\text{Ra}$ **C** ${}^{225}_{90}\text{Th}$ **D** ${}^{229}_{90}\text{Th}$

126 9702/12/O/N/17/Q40

Which combination of up (u) and down (d) quarks forms a neutron?

- A** u u u **B** u u d **C** u d d **D** d d d

127 9702/13/O/N/17/Q39

Each option shows the number of neutrons, protons and electrons in two atoms, some of which are ionised.

Which pair of atoms in the table are isotopes of the same element?

	neutron	proton	electron
A	20 20	17 19	20 20
B	28 28	23 24	23 24
C	34 36	29 29	29 28
D	40 42	32 31	32 32

128 9702/13/M/J/18/Q40

Which list contains only leptons?

- A** electron, neutrino, positron **C** electron, proton, neutron
B electron, neutrino, proton **D** neutrino, neutron, positron

129 9702/11/M/J/18/Q38

Which elementary particle is a lepton?

- A** proton **B** neutron **C** electron **D** quark

130 9702/11/M/J/18/Q39

How many down quarks are in a nucleus of hydrogen-3, ${}^3_1\text{H}$?

- A** 2 **B** 3 **C** 4 **D** 5

131 9702/11/M/J/18/Q40

What is the correct equation for β^+ decay?

- A** neutron $\rightarrow$ proton + electron + electron antineutrino
B neutron $\rightarrow$ proton + electron + electron neutrino
C proton $\rightarrow$ neutron + positron + electron antineutrino
D proton $\rightarrow$ neutron + positron + electron neutrino

132 9702/12/M/J/18/Q38

In the α -particle scattering experiment, a beam of α -particles is aimed at a thin gold foil. Most of the α -particles go straight through or are deflected by a small angle. A very small proportion are deflected through more than 90° , effectively rebounding towards the source of the α -particles.

Which conclusion about the structure of atoms **cannot** be drawn from this experiment alone?

- A** Most of the atom is empty space.
B Most of the mass of an atom is concentrated in the nucleus.
C The nucleus contains both protons and neutrons.
D The nucleus is charged.

133 9702/12/M/J/18/Q39

Radon-211, $^{211}_{86}\text{Rn}$, francium-210, $^{210}_{87}\text{Fr}$, and radium-212, $^{212}_{88}\text{Ra}$, are three nuclides.

How many neutrons does each nuclide have in its nucleus?

	radon-211	francium-210	radium-212
A	86	87	88
B	125	123	124
C	211	210	212
D	297	297	300

134 9702/12/M/J/18/Q40

A neutron is composed of one up (u) quark and two down (d) quarks. When the neutron decays to a proton, there is β -emission.

What is the change in the quark structure of the neutron due to the β -emission?

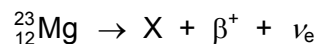
(The symbol for a neutrino is ν_e and for an antineutrino is $\bar{\nu}_e$.)

A $d \rightarrow u + \beta^- + \nu_e$ **C** $u \rightarrow d + \beta^+ + \nu_e$

B $d \rightarrow u + \beta^- + \bar{\nu}_e$ **D** $u \rightarrow d + \beta^+ + \bar{\nu}_e$

135 9702/13/M/J/18/Q39

A nucleus of magnesium-23 undergoes β^+ decay, as represented by the nuclear equation shown.



What is nucleus X?

A $^{22}_{11}\text{Na}$

B $^{22}_{13}\text{Al}$

C $^{23}_{11}\text{Na}$

D $^{23}_{13}\text{Al}$

136 9702/12/F/M/18/Q40

An isolated neutron decays to produce a proton, a β^- particle and an antineutrino.

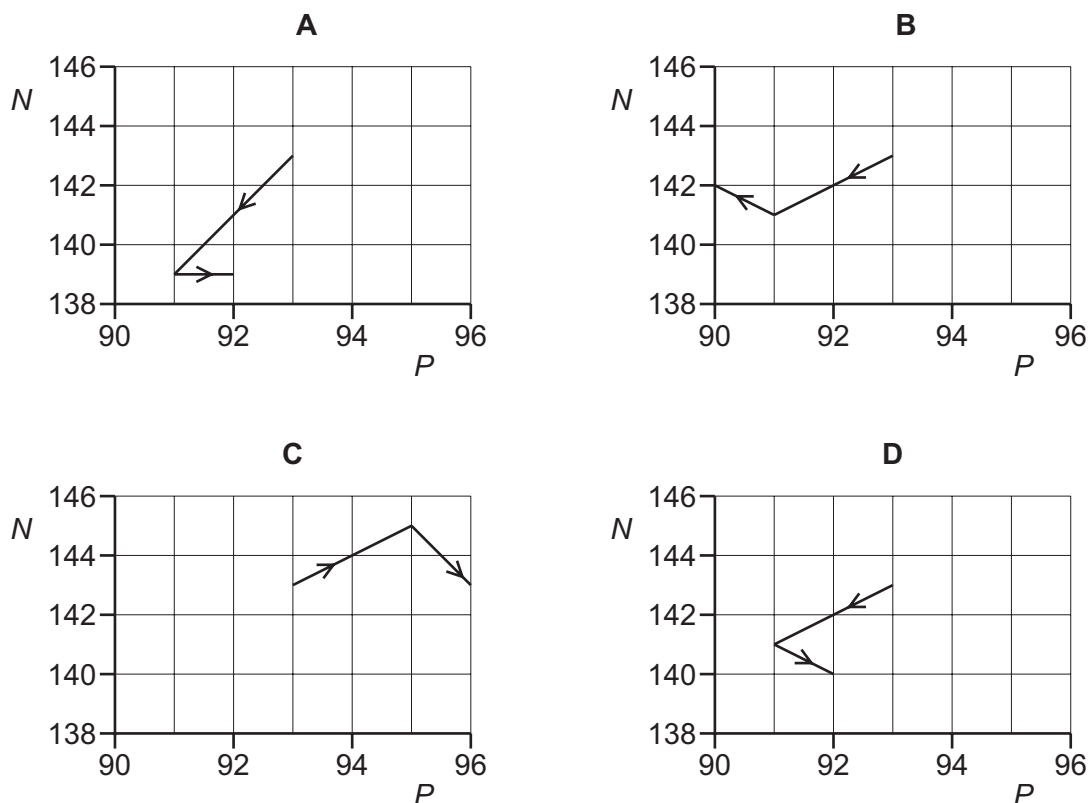
Which row gives the quark composition of the neutron and the proton and the type of force that gives rise to this reaction?

	quark composition		type of force
	neutron	proton	
A	down, down, up	down, up, up	strong interaction
B	down, down, up	down, up, up	weak interaction
C	down, up, up	down, down, up	strong interaction
D	down, up, up	down, down, up	weak interaction

137 9702/12/F/M/18/Q39

A nucleus of neptunium-236 contains 93 protons and 143 neutrons. This nucleus decays with the emission of an α -particle. The nucleus formed then emits a β^- particle.

Which diagram shows the changes in the number P of protons and the number N of neutrons in these nuclei?



138 9702/12/F/M/18/Q40

An isolated neutron decays to produce a proton, a β^- particle and an antineutrino.

Which row gives the quark composition of the neutron and the proton and the type of force that gives rise to this reaction?

	quark composition		type of force
	neutron	proton	
A	down, down, up	down, up, up	strong interaction
B	down, down, up	down, up, up	weak interaction
C	down, up, up	down, down, up	strong interaction
D	down, up, up	down, down, up	weak interaction

139 9702/11/O/N/18/Q39

A nucleus emits a β^- particle.

What is the change to the proton number and to the nucleon number of the nucleus?

	proton number	nucleon number
A	-1	+1
B	0	-1
C	+1	-1
D	+1	0

140 9702/11/O/N/18/Q40

How many up quarks and how many down quarks are in a nucleus of the nuclide $^{37}_{17}\text{Cl}$?

	up quarks	down quarks
A	51	60
B	54	57
C	57	54
D	60	51

141 9702/12/O/N/18/Q39

A proton in a nucleus undergoes β^+ decay. One of the products is a neutron.

What are the other products?

- A** an electron and a neutrino
- B** an electron and an antineutrino
- C** a positron and a neutrino
- D** a positron and an antineutrino

142 9702/12/O/N/18/Q40

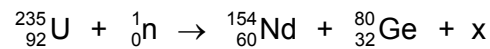
A certain type of hadron has zero charge. It is composed of a down quark, a strange quark and one other quark.

What could be the other quark?

- A** up
- B** down
- C** strange
- D** anti-strange

143 9702/13/O/N/18/Q38

A neutron collides with a nucleus of uranium-235. One possible nuclear reaction that results is represented by the equation



where x represents one or more particles.

What does x represent?

- A** one neutron
- B** two electrons
- C** two neutrons
- D** two protons

144 9702/13/O/N/18/Q39

A nucleus Q has the notation ${}_x^y\text{Q}$.

Which of the following is an isotope of Q?

- A** ${}_x^{y-1}\text{Q}$ **B** ${}_{x-1}^y\text{Q}$ **C** ${}_{x+1}^y\text{Q}$ **D** ${}_{x+1}^{y-1}\text{Q}$

145 9702/13/O/N/18/Q40

In β^- decay, a neutron inside a nucleus changes to a proton.

Which statement describes the quark composition of the nucleus during the decay?

- A** The number of down quarks decreases by one.
- B** The number of down quarks increases by one.
- C** The number of down quarks stays the same.
- D** The number of up quarks stays the same.

146 9702/11/M/J/19/Q39

Which statement about the alpha-particle scattering experiment provides evidence for the existence of the nucleus?

- A** A tiny proportion of the alpha-particles are deflected through large angles.
- B** Slower alpha-particles are deflected through larger angles.
- C** The kinetic energies of the deflected alpha-particles are unchanged.
- D** The number of alpha-particles deflected depends on the thickness of the foil.

147 9702/11/M/J/19/Q40

Some particles are a combination of three quarks.

Which combination of quarks would **not** result in a particle with a charge of either $+1.6 \times 10^{-19} \text{ C}$ or zero?

- A** up, down, down
- B** up, strange, strange
- C** up, up, down
- D** up, up, up

148 9702/12/M/J/19/Q38

A nucleus of francium-221 ($^{221}_{87}\text{Fr}$) decays into a nucleus of bismuth-209 ($^{209}_{83}\text{Bi}$) in several steps.

Which particles could be emitted?

- A** 2 α -particles and 4 β^- particles
- B** 2 α -particles and 4 β^+ particles
- C** 3 α -particles and 2 β^- particles
- D** 3 α -particles and 2 β^+ particles

149 9702/12/M/J/19/Q39

Which equation describes the changes to the quark composition of a nucleus and the lepton emission during the process of β^+ decay?

- A** down $\rightarrow$ up + positron + electron neutrino
- B** down $\rightarrow$ up + positron + electron antineutrino
- C** up $\rightarrow$ down + positron + electron neutrino
- D** up $\rightarrow$ down + positron + electron antineutrino

150 9702/12/M/J/19/Q40

There are protons, neutrons and electrons in the simple model of an atom.

To which class (group), hadron or lepton, do these particles belong?

	hadron	lepton
A	electron	proton and neutron
B	neutron	proton and electron
C	proton and electron	neutron
D	proton and neutron	electron

151 9702/13/M/J/19/Q39

What is a correct estimate of the order of magnitude of the diameter of a typical atomic nucleus?

- A** 10^{-14} m **B** 10^{-18} m **C** 10^{-22} m **D** 10^{-26} m

152 9702/13/M/J/19/Q40

Which statement describes β^- decay in terms of a simple quark model?

- A** A down quark changes to an up quark, and an electron and an electron antineutrino are emitted.
B A down quark changes to an up quark, and a positron and an electron neutrino are emitted.
C An up quark changes to a down quark, and an electron and an electron antineutrino are emitted.
D An up quark changes to a down quark, and a positron and an electron neutrino are emitted.

153 9702/11/O/N/19/Q39

An unstable nucleus of an element decays by emitting an α -particle or a β^- particle to become a nucleus of a different element. This nucleus is also unstable and also emits an α -particle or a β^- particle. The process continues until an isotope of the original element is produced.

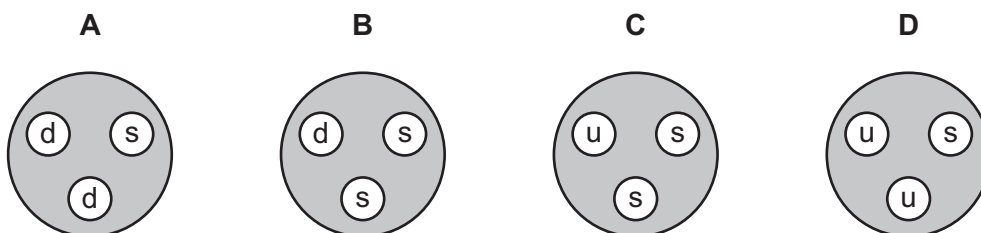
What is the minimum possible number of these particles emitted?

- A** 2 **B** 3 **C** 4 **D** 5

154 9702/11/O/N/19/Q40

The diagrams show the quark composition of four different hadrons. One of the hadrons is a Σ^+ particle. It has a charge of $+e$, where e is the elementary charge.

Which hadron could be the Σ^+ particle?



key
u = up quark
d = down quark
s = strange quark

155 9702/12/O/N/19/Q38

The table gives some data relating to four neutral (uncharged) atoms W, X, Y and Z.

	W	X	Y	Z
nucleon (mass) number	16	17	17	18
total number of particles (protons, neutrons and electrons) in the atom	24	26	25	28

Two of the atoms are isotopes of the same element.

What is the proton number of this element?

- A** 7 **B** 8 **C** 9 **D** 10

156 9702/12/O/N/19/Q39

What is **not** a fundamental particle?

- A** electron
B neutrino
C neutron
D positron

157 9702/12/O/N/19/Q40

An unstable nucleus decays and emits a β^- particle.

Which changes, if any, occur to the quark composition of the nucleus?

	quark changes	
	up quarks	down quarks
A	+1	0
B	+1	-1
C	-1	+1
D	0	+1

158 9702/13/O/N/19/Q39

A radioactive nucleus emits an α -particle or a β^- particle, creating a product nucleus.

Which decay could create the product nucleus stated?

	radioactive nucleus	decay	product nucleus
A	$^{226}_{88}\text{Ra}$	α	$^{224}_{86}\text{Rn}$
B	$^{238}_{92}\text{U}$	α	$^{242}_{94}\text{Pu}$
C	$^{228}_{88}\text{Ra}$	β^-	$^{228}_{87}\text{Fr}$
D	$^{231}_{90}\text{Th}$	β^-	$^{231}_{91}\text{Pa}$

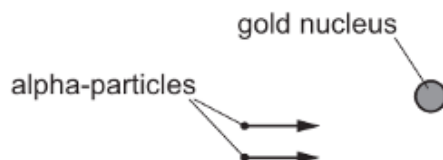
159 9702/13/O/N/19/Q40

Which statement is correct?

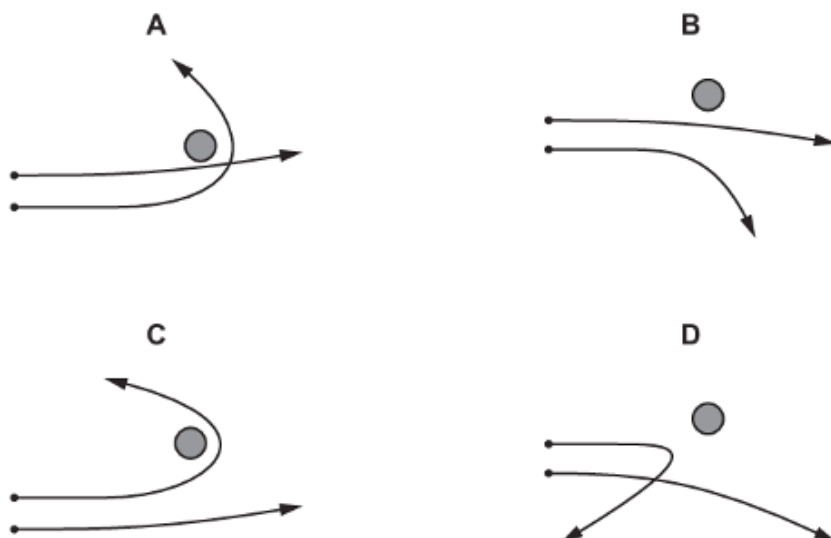
- A** Electrons and neutrinos are fundamental particles.
- B** Electrons and neutrinos are hadrons.
- C** Protons and neutrons are leptons.
- D** Protons and neutrons are quarks.

160. 9702/12/F/M/20/Q38

Two alpha-particles with the same kinetic energy are moving towards, and are then deflected by, a gold nucleus.

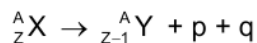


Which diagram could show the paths of the two alpha-particles?



161. 9702/12/F/M/20/Q39

The equation represents the decay of a nucleus X to a nucleus Y.



What are particles p and q?

	p	q
A	β^- particle	neutron
B	β^- particle	proton
C	β^+ particle	antineutrino
D	β^+ particle	neutrino

162. 9702/12/F/M/20/Q40

Which row gives the correct type and quark composition for the named particle?

	particle	type	quark composition
A	neutron	hadron	u u d
B	neutron	lepton	u d d
C	proton	hadron	u u d
D	proton	lepton	u d d

key

u = up quark

d = down quark

163. 9702/11/M/J/20/Q39

An element has two isotopic forms.

What are the nuclear arrangements of these two isotopes?

- A** They have different nucleon numbers and different proton numbers.
- B** They have different nucleon numbers but the same proton number.
- C** They have the same nucleon number and the same proton number.
- D** They have the same nucleon number but different proton numbers.

164. 9702/11/M/J/20/Q40

A hadron has a charge $+e$, where e is the elementary charge.

Which combination of up (u) and down (d) quarks could form this hadron?

- A** ddd
- B** udd
- C** uud
- D** uuu

165. 9702/12/M/J/20/Q39

An unstable nucleus goes through successive decays to become a final, stable nucleus.

The initial nucleus and the final nucleus are isotopes of each other.

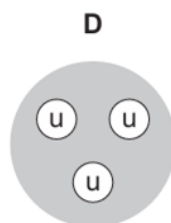
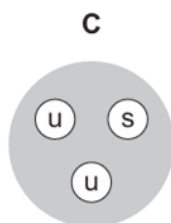
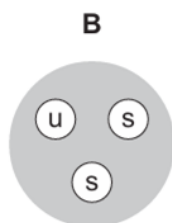
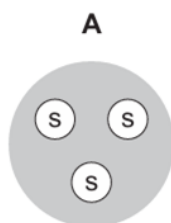
How many α and β^- particles could have been emitted during the decay sequence?

	particle	
	α	β^-
A	1	0
B	1	2
C	2	0
D	2	1

166. 9702/12/M/J/20/Q40

A hadron has a charge of $-e$ and is composed of three quarks.

What could be the quark composition of the hadron?



key

u = up quark
s = strange quark

167. 9702/13/M/J/20/Q39

Radiation from a radioactive source has a range of a few millimetres in air and can be deflected by an electric field.

Which type of radiation is being emitted?

- A** α -radiation
- B** β^- radiation
- C** β^+ radiation
- D** γ -rays

168. 9702/13/M/J/20/Q40

Which equation describes the process of β^+ decay?

key:

u = up quark

d = down quark

ν = (electron) neutrino

$\bar{\nu}$ = (electron) antineutrino

A $ddu \rightarrow uud + \beta^+ + \nu$

B $ddu \rightarrow uud + \beta^+ + \bar{\nu}$

C $uud \rightarrow ddu + \beta^+ + \nu$

D $uud \rightarrow ddu + \beta^+ + \bar{\nu}$

169. 9702/11/O/N/20/Q38

Which statement about two nuclei that are isotopes of the same element is correct?

A The nuclei each have the same acceleration when in the same uniform electric field.

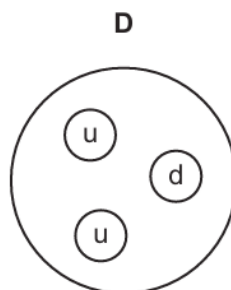
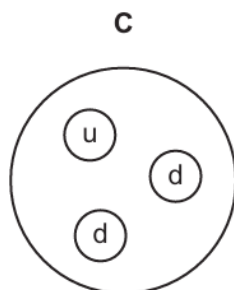
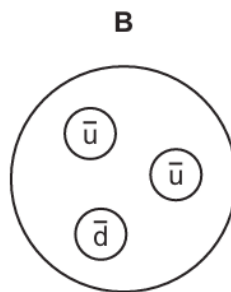
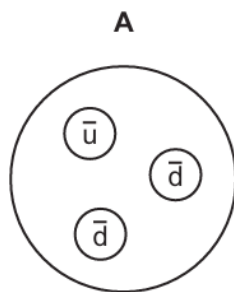
B The nuclei each have the same number of neutrons.

C The nuclei each have the same number of nucleons.

D Uncharged atoms containing the nuclei each have the same number of electrons.

170. 9702/11/O/N/20/Q40

Which diagram represents the quark composition of an antineutron?



key

u up quark

d down quark

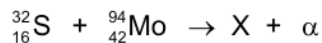
$\bar{u}$ up antiquark

$\bar{d}$ down antiquark

171. 9702/11/O/N/20/Q39

In a nuclear physics experiment, a nucleus of $^{32}_{16}\text{S}$ collides with a nucleus of $^{94}_{42}\text{Mo}$. The nuclei combine together and immediately emit a single alpha-particle.

The nuclear reaction is shown.



What is nucleus X?

A $^{122}_{56}\text{X}$

B $^{124}_{54}\text{X}$

C $^{126}_{58}\text{X}$

D $^{126}_{59}\text{X}$

172. 9702/12/O/N/20/Q39

What is a conclusion from the alpha-particle scattering experiment?

A Protons and electrons have equal but opposite charges.

B Protons have a much larger mass than electrons.

C The nucleus contains most of the mass of the atom.

D The nucleus of an atom contains protons and neutrons.

173. 9702/12/O/N/20/Q40

Which particle is a hadron?

A electron

B neutrino

C positron

D proton

174. 9702/13/O/N/20/Q39

The table contains data for four different nuclei P, Q, R and S.

nucleus	number of neutrons	nucleon number
P	5	10
Q	6	10
R	6	14
S	8	16

Which two nuclei are isotopes of the same element?

A P and Q

B P and S

C Q and R

D R and S

175. 9702/13/O/N/20/Q40

Which particle is a fundamental particle?

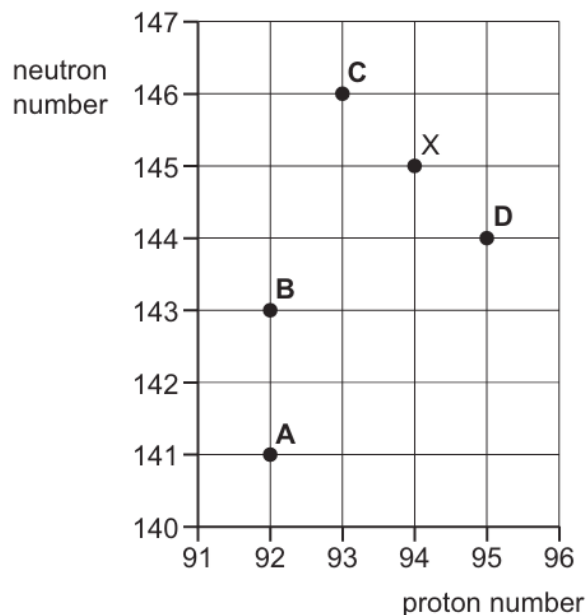
- A alpha-particle
- B electron
- C neutron
- D proton

176. 9702/12/F/M/21/Q39

The figure shows part of a chart of nuclides where neutron number is plotted against proton number.

An unstable nuclide X decays by emitting an α -particle.

Which nuclide is formed by the decay of nuclide X?

**177. 9702/12/F/M/21/Q40**

The nuclei of common isotopes of hydrogen, helium, lithium and beryllium are shown.

Which nucleus contains equal numbers of up and down quarks?

- A ${}^1_1\text{H}$
- B ${}^4_2\text{He}$
- C ${}^7_3\text{Li}$
- D ${}^9_4\text{Be}$

178. 9702/11/M/J/21/Q39

A nucleus of uranium, ${}^{235}_{92}\text{U}$, undergoes a series of decays. During the series of decays, two α -particles and one β^- particle are emitted.

As a result, a nucleus of actinium, Ac, is formed.

What is the correct notation for the nuclide of actinium that is formed?

- A ${}^{227}_{87}\text{Ac}$
- B ${}^{227}_{89}\text{Ac}$
- C ${}^{231}_{87}\text{Ac}$
- D ${}^{231}_{89}\text{Ac}$

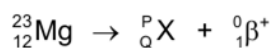
179. 9702/11/M/J/21/Q40

Which particle is a fundamental particle?

- A electron
- B hadron
- C neutron
- D proton

180. 9702/12/M/J/21/Q39

A nucleus of magnesium decays into a nucleus X by emitting a β^+ particle. The decay is represented by the equation shown.



What are the values of P and Q?

	P	Q
A	22	11
B	22	13
C	23	11
D	23	13

181. 9702/12/M/J/21/Q40

In β^- decay, a neutron inside a nucleus changes to a proton.

Which statement describes the quark composition of the nucleus during the decay?

- A The number of down quarks decreases by one.
- B The number of down quarks increases by one.
- C The number of down quarks stays the same.
- D The number of up quarks stays the same.

182. 9702/13/M/J/21/Q39

A beam of α -particles is incident on a thin gold foil. One α -particle collides head-on with a gold nucleus and is deflected back along its original path.

Which statement could explain why the recoil speed of the gold nucleus is small compared with the recoil speed of the α -particle?

- A Most α -particles are only slightly deflected as they pass through the gold foil.
- B The α -particle and the gold nucleus repel each other.
- C The mass of the gold nucleus is much greater than the mass of the α -particle.
- D The momentum of the α -particle decreases as it approaches the gold nucleus.

183. 9702/13/M/J/21/Q40

A hadron is composed of three quarks. The hadron has a charge.

What is a possible quark composition of the hadron?

- A down, down, up
- B down, up, strange
- C up, strange, strange
- D up, up, strange

184. 9702/11/O/N/21/Q39

A uranium-238 nucleus, $^{238}_{92}\text{U}$, undergoes a series of nuclear decays to form uranium-234, $^{234}_{92}\text{U}$.

Which series of decays could give this result?

- A emission of four β^- particles
- B emission of four γ -rays
- C emission of one α -particle and two β^- particles
- D emission of two α -particles and eight β^- particles

185. 9702/11/O/N/21/Q40

Which combination of up (u) and down (d) quarks forms a proton?

- A u u u B u u d C u d d D d d d

186. 9702/12/O/N/21/Q39

An unstable nucleus decays by emitting a β^+ particle.

Which statement is correct?

- A An antineutrino is also emitted.
- B A neutron changes into a proton.
- C Mass–energy is conserved.
- D The nucleon number is not conserved.

187. 9702/12/O/N/21/Q40

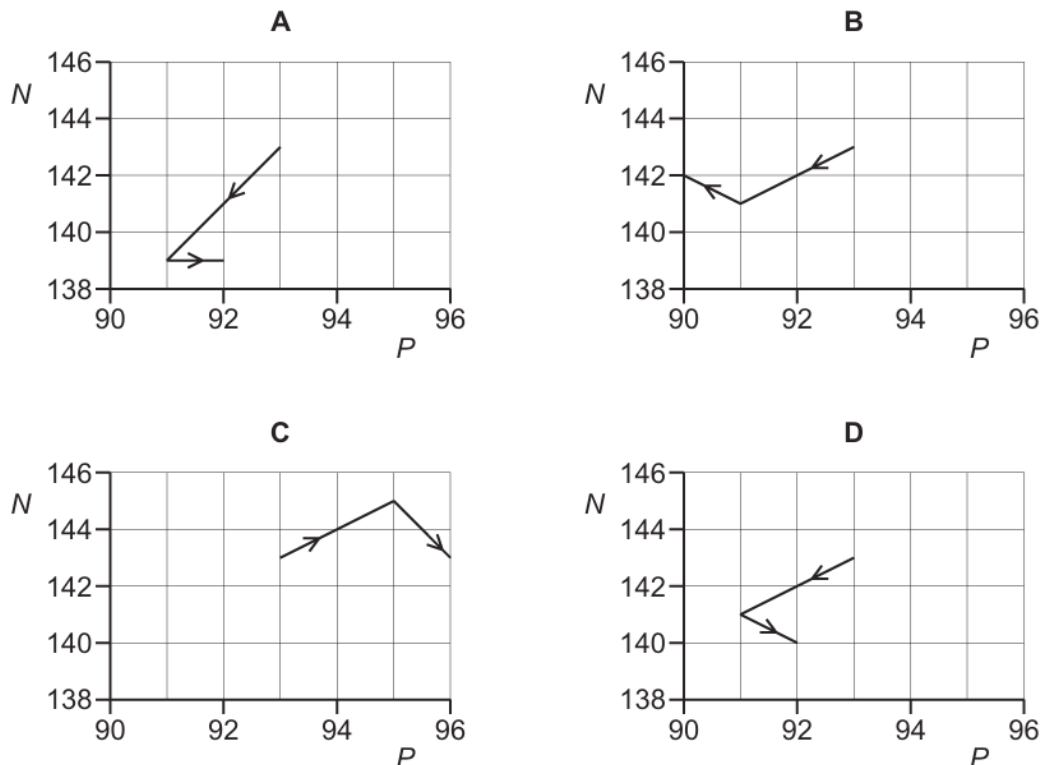
Which statement is **not** correct?

- A An antineutrino is a fundamental particle.
- B An electron is made up of a quark and an antiquark.
- C A neutrino is a lepton.
- D A neutron is composed of three quarks.

188. 9702/13/O/N/21/Q39

A nucleus of neptunium-236 contains 93 protons and 143 neutrons. This nucleus decays with the emission of an α -particle. The nucleus formed then emits a β^- particle.

Which diagram shows the changes in the number P of protons and the number N of neutrons in these nuclei?



189. 9702/13/O/N/21/Q40

Which combination of quarks could **not** be the quark composition of the hadron shown?

p = proton

n = neutron

Σ^- = sigma particle of charge $-e$

Σ^+ = sigma particle of charge $+e$

u = up quark

d = down quark

s = strange quark

	hadron	quark composition
A	Σ^-	dds
B	Σ^+	uds
C	p	uud
D	n	udd

190. 9702/13/O/N/21/Q38

When α -particles are fired at a thin metal foil, most of the particles pass straight through but a few are deflected by a large angle.

Which change would increase the **proportion** of α -particles deflected by a large angle?

- A** using α -particles with greater kinetic energy
- B** using a double thickness foil
- C** using a foil made of a metal with fewer protons in its nuclei
- D** using a source emitting more α -particles per unit time

191. 9702/12/F/M/22/Q38

When α -particles are directed at gold leaf:

- 1 almost all α -particles pass through without deflection
- 2 a few α -particles are deviated through large angles.

What are the reasons for these effects?

	1	2
A	most α -particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α -particles bounce back from the gold surface
B	most α -particles miss all gold atoms	a few α -particles bounce off gold atoms
C	the gold nucleus is very small so most α -particles miss all nuclei	occasionally the path of an α -particle is close to a nucleus
D	the positive charge in an atom is not concentrated enough to deflect an α -particle	occasionally an α -particle experiences many small deflections in the same direction

192. 9702/12/F/M/22/Q39

A nucleus X is radioactive and decays into a nucleus Y.

X and Y are isotopes of the same element.

Which combination of particles could have been emitted during the decay process?

- A** 1 α -particle and 1 β^- particle
- B** 1 α -particle and 2 β^- particles
- C** 2 α -particles and 1 β^- particle
- D** 2 α -particles and 2 β^- particles

193. 9702/12/F/M/22/Q40

A positively charged meson consists of a quark and an antiquark.

What could be the quark and antiquark?

- A charm and antiup
- B down and antitop
- C strange and antibottom
- D up and antistrange

194. 9702/11/M/J/22/Q38

Carbon-14 decays into nitrogen-14 by emitting a β^- particle.

Which statement explains why the β^- particles are emitted with a range of different kinetic energies?

- A The carbon-14 nuclei have slightly different masses.
- B The emitted β^- particles have a range of different masses.
- C The energy released in the decay process is different for each carbon-14 nucleus that decays.
- D The energy released in the decay process is shared between the nitrogen-14 nucleus, a β^- particle and an antineutrino.

195. 9702/11/M/J/22/Q39

A nucleus of a radioactive element emits an α -particle, then a β^- particle and then another β^- particle.

Which statement describes the final element that is produced?

- A It is a different element of higher proton number than the original element.
- B It is a different element of lower nucleon number than the original element.
- C It is an isotope of the original element.
- D It is the same element as the original element but with a different proton number.

196. 9702/11/M/J/22/Q40

How many hadrons, baryons and mesons are there in a nucleus of ${}^9_4\text{Be}$?

	hadrons	baryons	mesons
A	9	4	5
B	9	5	4
C	9	9	0
D	13	9	0

197. 9702/12/M/J/22/Q38

In an experiment on α -particle scattering, α -particles are directed at a thin gold foil. Most of the α -particles pass straight through the foil or are deflected by a small angle. A small number of α -particles are deflected by a large angle.

Which statement **cannot** be deduced from this experiment?

- A Atoms are mostly empty space.
- B Most of the mass of an atom is concentrated in the nucleus.
- C The nucleus of an atom contains protons.
- D The nucleus of an atom is small compared to the size of an atom.

198. 9702/12/M/J/22/Q39

Four nuclei are represented below.



Which statement about these nuclei is correct?

- A An uncharged atom of element Q has 24 orbital electrons.
- B Nuclei G and M are isotopes of the same element.
- C When E absorbs a neutron and then emits an α -particle, E transforms into M.
- D When M emits a β^- particle, M transforms into Q.

199. 9702/12/M/J/22/Q40

A neutron is composed of one up (u) quark and two down (d) quarks. When a neutron decays to a proton, a beta-particle is emitted.

What is the change in the quark structure of the neutron due to the emission of the beta-particle?

(The symbol for a neutrino is ν_e and for an antineutrino is $\bar{\nu}_e$.)

- A $d \rightarrow u + \beta^- + \nu_e$
- B $d \rightarrow u + \beta^- + \bar{\nu}_e$
- C $u \rightarrow d + \beta^+ + \nu_e$
- D $u \rightarrow d + \beta^+ + \bar{\nu}_e$

200. 9702/13/M/J/22/Q40

Which statement is **not** correct?

- A A meson consists of three quarks.
- B A proton is a baryon.
- C A quark is a fundamental particle.
- D There are six flavours (types) of quark.

201. 9702/13/M/J/22/Q39

An actinium nucleus has a nucleon number of 227 and a proton number of 89. It decays to form a radium nucleus, emitting a β^- particle and an α -particle in the process.

What are the nucleon number and the proton number of this radium nucleus?

	nucleon number	proton number
A	223	87
B	223	88
C	224	87
D	225	86

202. 9702/11/O/N/22/Q38

An α -particle passes close to a gold nucleus and is deflected through an angle greater than 90° .

Which property of the α -particle changes as a result of the deflection?

- A** charge
- B** momentum
- C** nucleon number
- D** proton number

203. 9702/11/O/N/22/Q39

A nucleus of ${}_{92}^{238}\text{U}$ decays in stages by emitting α -particles and β^- particles, eventually forming a nucleus of ${}_{82}^{206}\text{Pb}$.

How many α -particles and how many β^- particles are emitted during the decay chain?

	α -particles	β^- particles
A	8	6
B	8	10
C	16	6
D	16	22

204. 9702/11/O/N/22/Q40

Which list of particles contains **only** fundamental particles?

- A** antineutrino, beta, meson
- B** baryon, neutrino, positron
- C** electron, neutrino, alpha
- D** lepton, quark, positron

205. 9702/12/O/N/22/Q38

In the α -particle scattering experiment, a beam of α -particles is aimed at a thin gold foil. Most of the α -particles go straight through or are deflected by a small angle. A very small proportion are deflected by more than 90° , effectively rebounding towards the source of the α -particles.

Which conclusion about the structure of atoms **cannot** be drawn from this experiment alone?

- A Most of the atom is empty space.
- B Most of the mass of an atom is concentrated in the nucleus.
- C The nucleus contains both protons and neutrons.
- D The nucleus is charged.

206. 9702/12/O/N/22/Q39

Which two particles have opposite charges?

- A alpha-particle and helium nucleus
- B antiproton and beta-plus particle
- C beta-minus particle and electron
- D positron and proton

207. 9702/12/O/N/22/Q40

Which particle is a lepton?

- A meson
- B positron
- C proton
- D quark

208. 9702/13/O/N/22/Q38

Which statement about the alpha-particle scattering experiment provides evidence for the existence of the nucleus?

- A A tiny proportion of the alpha-particles are deflected through large angles.
- B Slower alpha-particles are deflected through larger angles.
- C The kinetic energies of the deflected alpha-particles are unchanged.
- D The number of alpha-particles deflected depends on the thickness of the foil.

209. 9702/13/O/N/22/Q40

A particle consists of two up quarks and a charm quark.

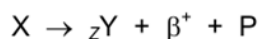
What is the charge of this particle, expressed in terms of the elementary charge e ?

- A $-2e$
- B $-e$
- C $+e$
- D $+2e$

210. 9702/13/O/N/22/O39

A nucleus X undergoes β^+ decay.

The products are a nucleus Y with proton number Z, a β^+ particle and another particle P.



What is particle P and what is the proton number of nucleus X?

	P	proton number of nucleus X
A	antineutrino	$Z - 1$
B	antineutrino	$Z + 1$
C	neutrino	$Z - 1$
D	neutrino	$Z + 1$

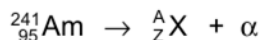
211. 9702/12/F/M/23/Q38

What is a conclusion from the alpha-particle scattering experiment?

- A** Protons and electrons have equal but opposite charges.
- B** Protons have a much larger mass than electrons.
- C** The nucleus contains most of the mass of the atom.
- D** The nucleus of an atom contains protons and neutrons.

212. 9702/12/F/M/23/Q39

Americium-241 is a radioactive nuclide used in smoke detectors. It undergoes α -decay to form nuclide X. This decay may be represented by the equation shown.



What are the values of A and Z?

	A	Z
A	237	93
B	239	91
C	241	94
D	241	96

213. 9702/12/F/M/23/Q40

A top quark has a charge of $+\frac{2}{3}e$, where e is the elementary charge.

What is the charge of an anti top quark?

- A** $-\frac{2}{3}e$
- B** $-\frac{1}{3}e$
- C** $+\frac{1}{3}e$
- D** $+\frac{2}{3}e$

214. 9702/11/M/J/23/Q38

A proton has mass m_p and charge $+e$.

What are the mass and charge of an antiproton?

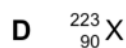
	mass	charge
A	$-m_p$	$+e$
B	$-m_p$	$-e$
C	m_p	$+e$
D	m_p	$-e$

215. 9702/11/M/J/23/Q39

A uranium nucleus has 92 protons and 143 neutrons.

The nucleus emits a total of 3 α -particles and 4 β^- particles to form nucleus X.

How can nucleus X be represented?

**216. 9702/11/M/J/23/Q40**

Which statement about a proton is correct?

- A** A proton is a baryon but not a meson.
- B** A proton is a meson but not a hadron.
- C** A proton is both a hadron and a meson.
- D** A proton is both a meson and a baryon.

217. 9702/12/M/J/23/Q38

What are isotopes?

- A** nuclei of different elements with the same number of neutrons
- B** nuclei of different elements with the same number of nucleons
- C** nuclei of the same element with different numbers of neutrons
- D** nuclei of the same element with different numbers of protons

218. 9702/12/M/J/23/Q39

A neutron ${}_0^1\text{n}$ is fired at a ${}_{92}^{235}\text{U}$ nucleus. The neutron is absorbed by the nucleus which then splits to form nuclei of ${}_{56}^{141}\text{Ba}$ and ${}_{36}^{92}\text{Kr}$.

What is the number of free neutrons emitted when the ${}_{92}^{235}\text{U}$ nucleus splits?

- A** 0
- B** 1
- C** 2
- D** 3

219. 9702/12/M/J/23/Q40

A π^+ meson has a charge of $+e$, where e is the elementary charge. It consists of an up quark and one other quark.

What could be the other quark in the π^+ meson?

- A** anti-down
- B** anti-up
- C** bottom
- D** charm

220. 9702/13/M/J/23/Q38

The table contains data for four different nuclei P, Q, R and S.

nucleus	number of neutrons	nucleon number
P	5	10
Q	6	10
R	6	14
S	8	16

Which two nuclei are isotopes of the same element?

- A** P and Q
- B** P and S
- C** Q and R
- D** R and S

221. 9702/13/M/J/23/Q39

Which statement about β^- decay is correct?

- A** A neutron changes to a proton in the nucleus and an electron and an antineutrino are emitted.
- B** A neutron changes to a proton in the nucleus and an electron and a neutrino are emitted.
- C** A proton changes to a neutron in the nucleus and an electron and an antineutrino are emitted.
- D** A proton changes to a neutron in the nucleus and an electron and a neutrino are emitted.

222. 9702/13/M/J/23/Q40

What is the quark composition of a hydrogen-3 nucleus, ${}^3_1\text{H}$?

	number of quarks	
	up	down
A	4	5
B	5	4
C	5	7
D	7	5

223. 9702/11/O/N/23/Q39

Which nuclide is formed when $^{10}_6\text{C}$ undergoes β^+ decay?

- A** $^{11}_6\text{C}$ **B** ^9_6C **C** $^{10}_5\text{B}$ **D** $^{10}_7\text{N}$

224. 9702/11/O/N/23/Q40

A particular hadron is composed of three quarks and has zero charge.

What is a possible quark composition of the hadron?

- A** down, down, strange
B up, down, strange
C up, up, down
D up, up, strange

225. 9702/12/O/N/23/Q38

Two nuclides are different isotopes of the same element.

Which statement about the nuclides is correct?

- A** Neutral atoms of the nuclides have the same number of electrons.
- B** Nuclei of the nuclides have different numbers of protons.
- C** Nuclei of the nuclides have the same number of nucleons.
- D** Nuclei of the nuclides have the same number of neutrons.

226. 9702/12/O/N/23/Q39

The charge-to-mass ratio r of a particle is given by the equation shown.

$$r = \frac{\text{charge on particle}}{\text{mass of particle}}$$

The value of r is determined for an α -particle, a β^+ particle and a proton p .

Which list shows the particles in order of increasing magnitude of r from left to right?

- A** $\alpha \rightarrow \beta^+ \rightarrow p$
B $\alpha \rightarrow p \rightarrow \beta^+$
C $p \rightarrow \alpha \rightarrow \beta^+$
D $p \rightarrow \beta^+ \rightarrow \alpha$

227. 9702/12/O/N/23/Q40

Which combination of up (u) and down (d) quarks forms a neutron?

- A** uuu **B** uud **C** udd **D** ddd

228. 9702/13/O/N/23/Q38

The table shows the number of nucleons and the total number of particles (protons, neutrons and electrons) in neutral atoms of four nuclides W, X, Y and Z.

	number of nucleons	total number of particles
W	19	30
X	19	31
Y	21	32
Z	22	31

Which two nuclides are isotopes of each other?

- A** W and X **B** W and Y **C** X and Z **D** Y and Z

229. 9702/13/O/N/23/Q39

When a sample of a radioactive isotope decays by α -particle emission, the α -particles emitted have a single discrete energy.

When a sample of a radioactive isotope decays by β^- particle emission, the β^- particles emitted have a continuous range of energies.

What is the explanation for this?

- A** An antineutrino is emitted with a β^- particle but not with an α -particle.
B An antineutrino is emitted with an α -particle but not with a β^- particle.
C The α -particles have much more energy than the β^- particles.
D The β^- particles have much more energy than the α -particles.

230. 9702/13/O/N/23/Q40

Some particles are a combination of three quarks.

Which combination of quarks does **not** result in a particle with a charge of either $+1.6 \times 10^{-19} \text{ C}$ or zero?

- A** up, down, down
B up, strange, strange
C up, up, down
D up, up, up